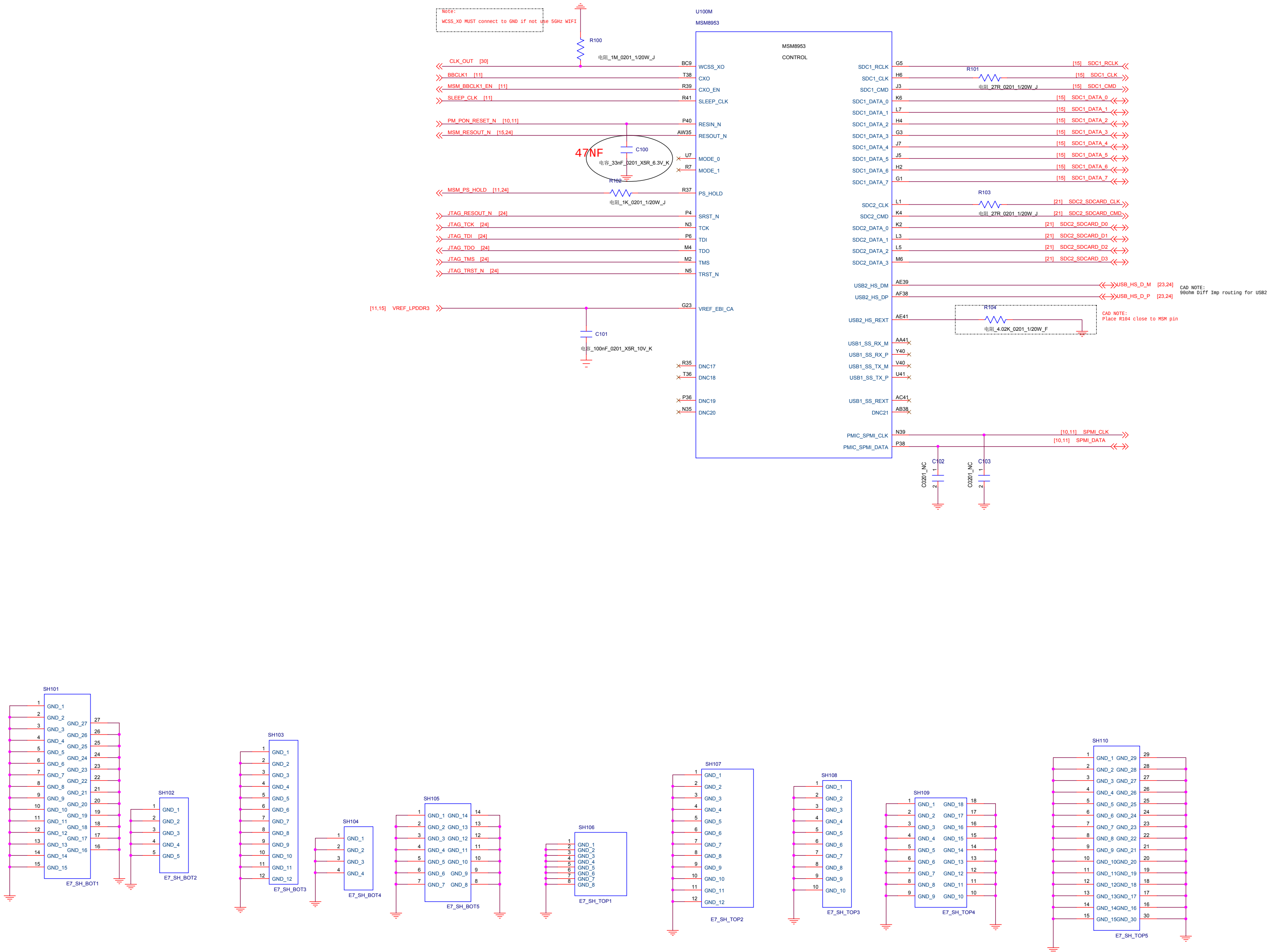
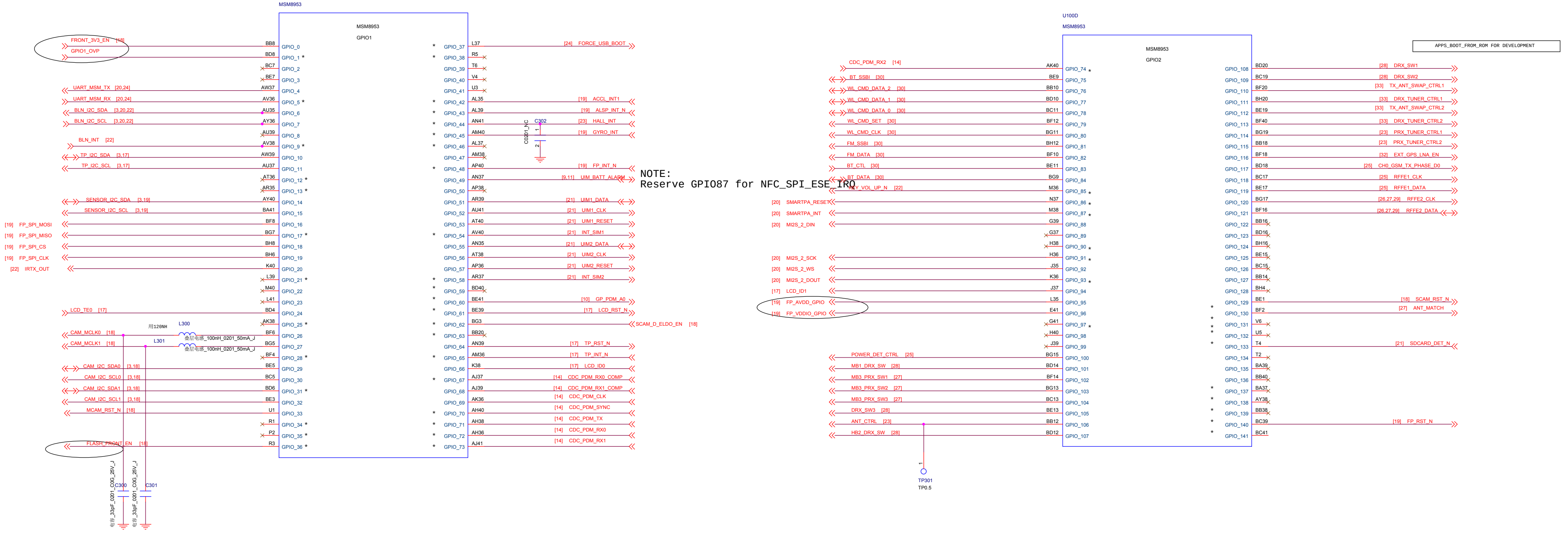


Title	E7_MB_V1
Size D	Document Number <Doc> MSM8953 Control
Rev	V1.0
Date:	Wednesday, August 09, 2017
Sheet	1 of 34

Title E7_MB_V1		
Size D	Document Number <Doc> MSM8953 EBI	Rev V1.0
Date: Wednesday, August 09, 2017	Sheet 2 of 34	

NOTE:
Asterisks (*) indicate modem power management (MPM) wake-up pins

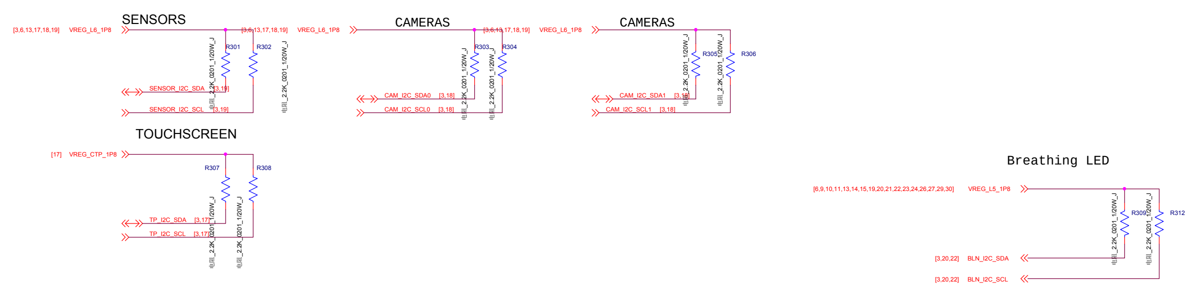


GPIO_37	FORCED_USB_BOOT
GPIO_106	WDOG_DISABLE
GPIO_108	APPS_BOOT_FROM_ROM

BOOT_CONFIG[3:1]	BOOT_CONFIG
0b000	SDC1 -> SDC2 -> USB2.0
0b001	SDC2 -> SDC1-> USB2.0
0b010	SDC1-> USB2.0
0b011	USB2.0

Default Boot Config (0b000) is SDC1(eMMC)

I2C PULL-UP RESISTORS

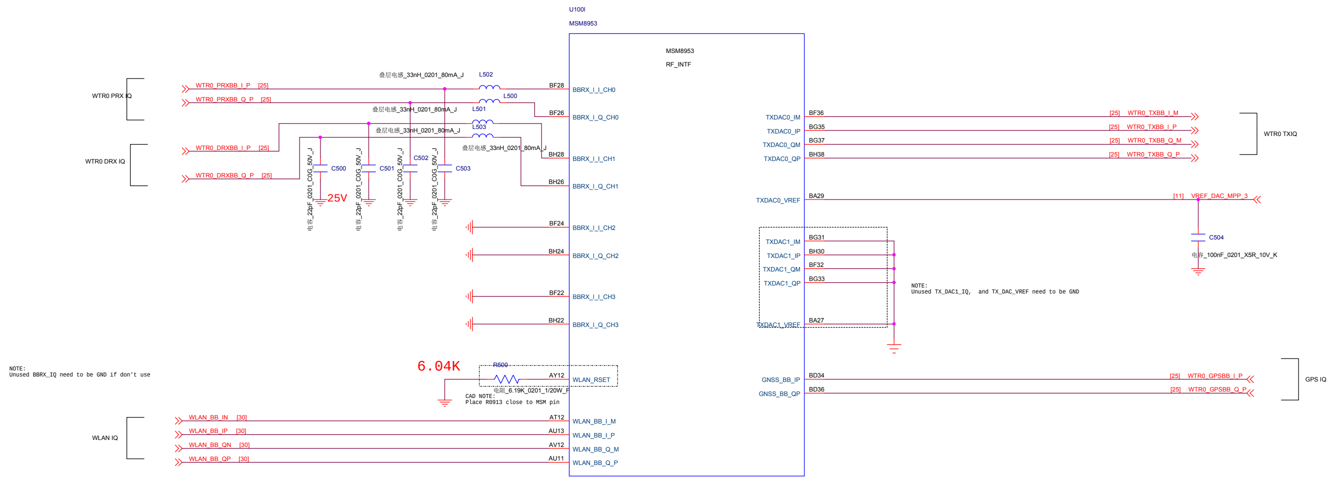


NOTE:
Ensure SW sets these GPIOs (Sensor, CTP and
Camera I2C bus) to inout pull down when the peripherals
are powered off to eliminate leakage.

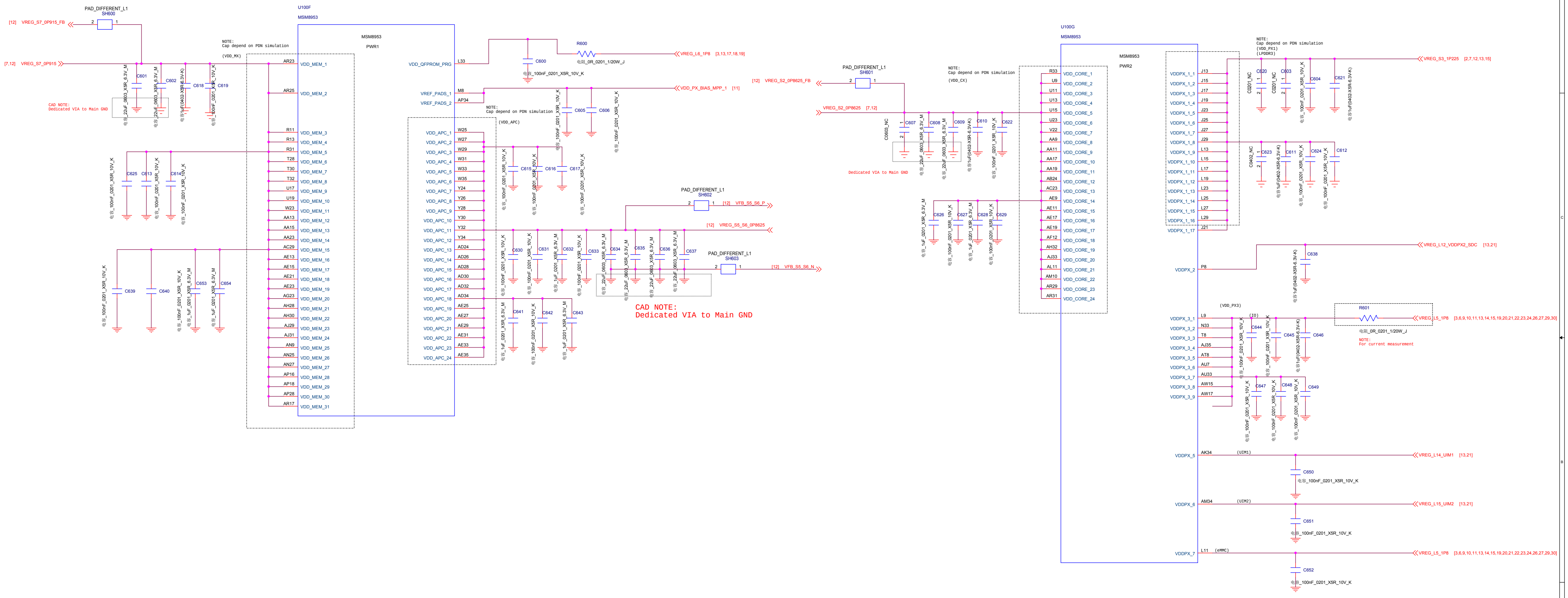
Title	E7_MB_V1
Size D	Document Number <Doc> MSM8953 GPIO — Rev V1.0
Date:	Wednesday, August 09, 2017 — Sheet 3 of 34

CSI Pin Name	CSI DPHY 4-lane	CSI DPHY 2+1 Mode	CSI CPHY 3Phase Mode
CSI0_3PHASE_PIN0	CSI0_CLKP	CSI0_2LANE_CLKP	NC
CSI0_3PHASE_PIN1	CSI0_CLKN	CSI0_2LANE_CLKN	CSI0_TRI0_A
CSI0_3PHASE_PIN2	CSI0_DP0	CSI0_2LANE_DP0	CSI0_TRI0_B
CSI0_3PHASE_PIN3	CSI0_DN0	CSI0_2LANE_DN0	CSI0_TRI0_C
CSI0_3PHASE_PIN4	CSI0_DP1	CSI0_2LANE_DP1	CSI0_TRI1_A
CSI0_3PHASE_PIN5	CSI0_DN1	CSI0_2LANE_DN1	CSI0_TRI1_B
CSI0_3PHASE_PIN6	CSI0_DP2	CSI0_1LANE_DP0	CSI0_TRI1_C
CSI0_3PHASE_PIN7	CSI0_DN2	CSI0_1LANE_DN0	CSI0_TRI2_A
CSI0_3PHASE_PIN8	CSI0_DP3	CSI0_1LANE_CLKP	CSI0_TRI2_B
CSI0_3PHASE_PIN9	CSI0_DN3	CSI0_1LANE_CLKN	CSI0_TRI2_C
CSI1_3PHASE_PIN0	CSI1_CLKP	CSI1_2LANE_CLKP	NC
CSI1_3PHASE_PIN1	CSI1_CLKN	CSI1_2LANE_CLKN	CSI1_TRI0_A
CSI1_3PHASE_PIN2	CSI1_DP0	CSI1_2LANE_DP0	CSI1_TRI0_B
CSI1_3PHASE_PIN3	CSI1_DN0	CSI1_2LANE_DN0	CSI1_TRI0_C
CSI1_3PHASE_PIN4	CSI1_DP1	CSI1_2LANE_DP1	CSI1_TRI1_A
CSI1_3PHASE_PIN5	CSI1_DN1	CSI1_2LANE_DN1	CSI1_TRI1_B
CSI1_3PHASE_PIN6	CSI1_DP2	CSI1_1LANE_DP0	CSI1_TRI1_C
CSI1_3PHASE_PIN7	CSI1_DN2	CSI1_1LANE_DN0	CSI1_TRI2_A
CSI1_3PHASE_PIN8	CSI1_DP3	CSI1_1LANE_CLKP	CSI1_TRI2_B
CSI1_3PHASE_PIN9	CSI1_DN3	CSI1_1LANE_CLKN	CSI1_TRI2_C
CSI2_3PHASE_PIN0	CSI2_CLKP	CSI2_2LANE_CLKP	NC
CSI2_3PHASE_PIN1	CSI2_CLKN	CSI2_2LANE_CLKN	CSI2_TRI0_A
CSI2_3PHASE_PIN2	CSI2_DP0	CSI2_2LANE_DP0	CSI2_TRI0_B
CSI2_3PHASE_PIN3	CSI2_DN0	CSI2_2LANE_DN0	CSI2_TRI0_C
CSI2_3PHASE_PIN4	CSI2_DP1	CSI2_2LANE_DP1	CSI2_TRI1_A
CSI2_3PHASE_PIN5	CSI2_DN1	CSI2_2LANE_DN1	CSI2_TRI1_B
CSI2_3PHASE_PIN6	CSI2_DP2	CSI2_1LANE_DP0	CSI2_TRI1_C
CSI2_3PHASE_PIN7	CSI2_DN2	CSI2_1LANE_DN0	CSI2_TRI2_A
CSI2_3PHASE_PIN8	CSI2_DP3	CSI2_1LANE_CLKP	CSI2_TRI2_B
CSI2_3PHASE_PIN9	CSI2_DN3	CSI2_1LANE_CLKN	CSI2_TRI2_C

Title: E7_MB_V1
Size: D
Document Number: <Doc> MSM8953 MIPI
Rev: V1.0
Date: Wednesday, August 09, 2017

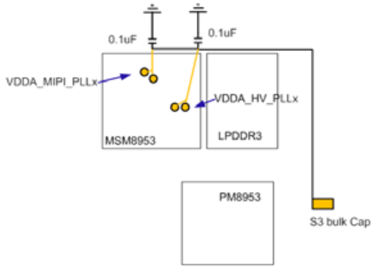


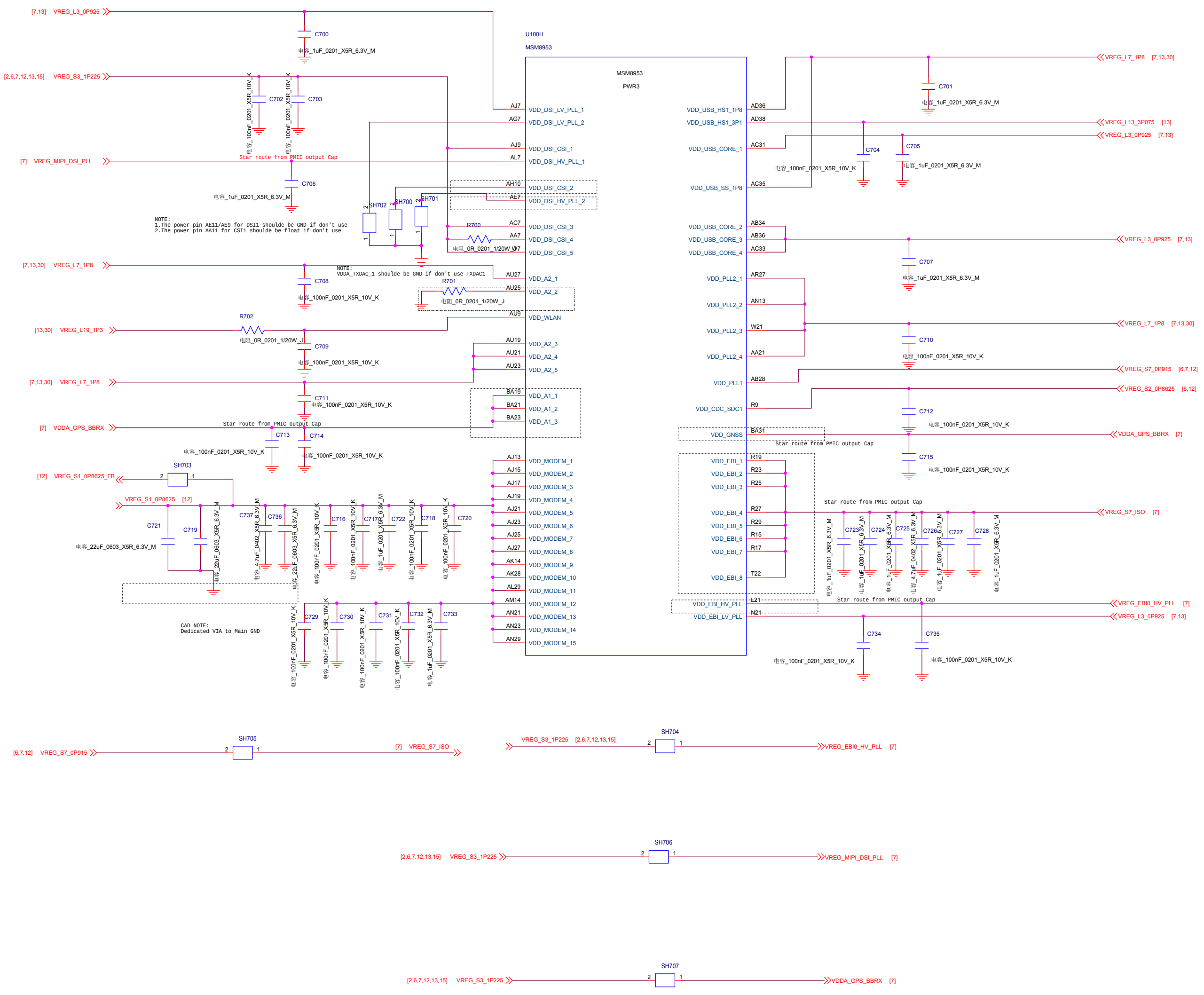
Title	E7_MB_V1
Size D	Document Number <Doc> MSM8953 RF	Rev V1.0
Date:	Wednesday, August 09, 2017	Sheet 5 of 34



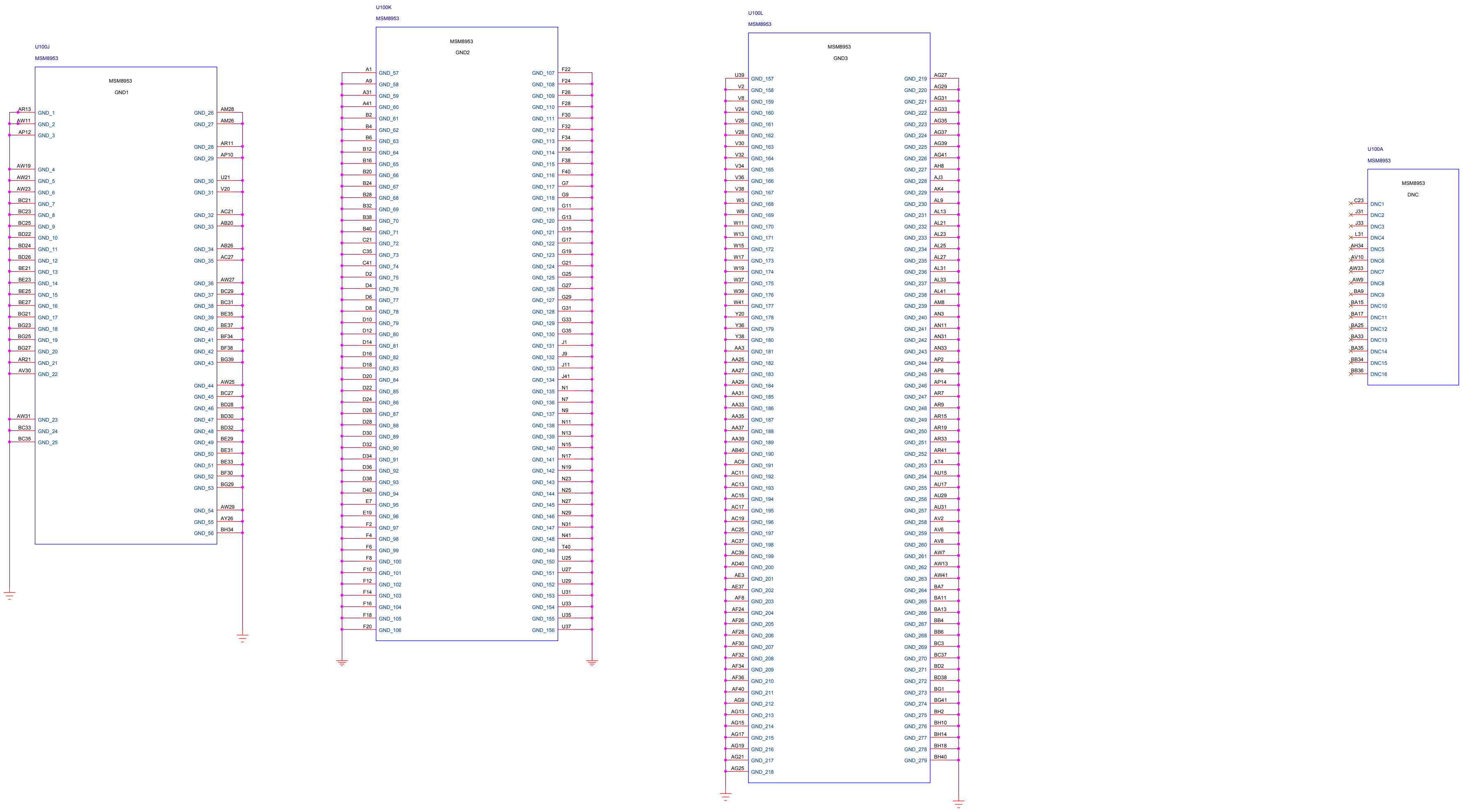
Title	E7_MB_V1	
Size D	Document Number <Doc> MSM8953 POWER1	Rev V1.0
Date: Wednesday, August 09, 2017	Sheet 6 of 34	

CAD NOTE:
About star route from PMIC output Cap:



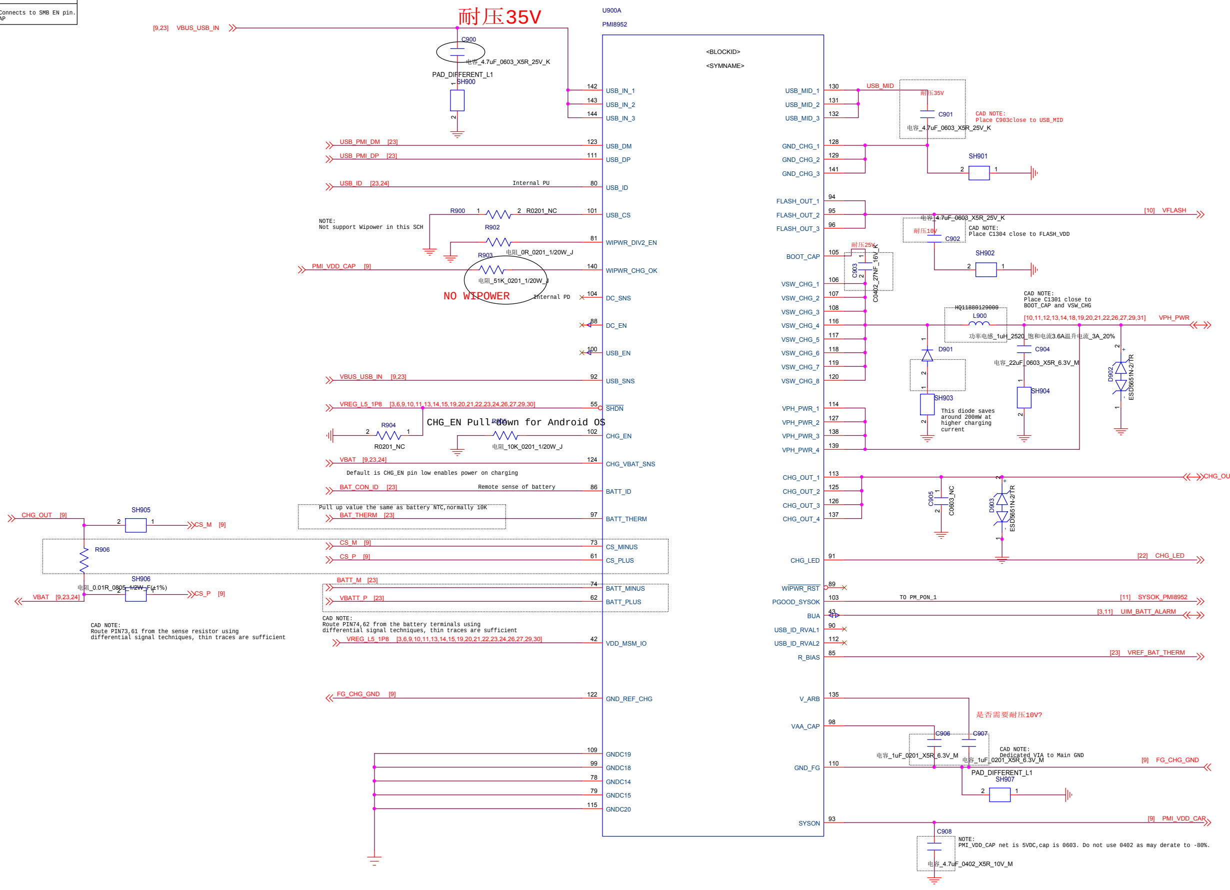


Title	E7_MB_V1		
Size D	Document Number <Doc> MSM8953 POWER2		Rev V1.0
Date:	Wednesday, August 09, 2017	Sheet 7 of 34	

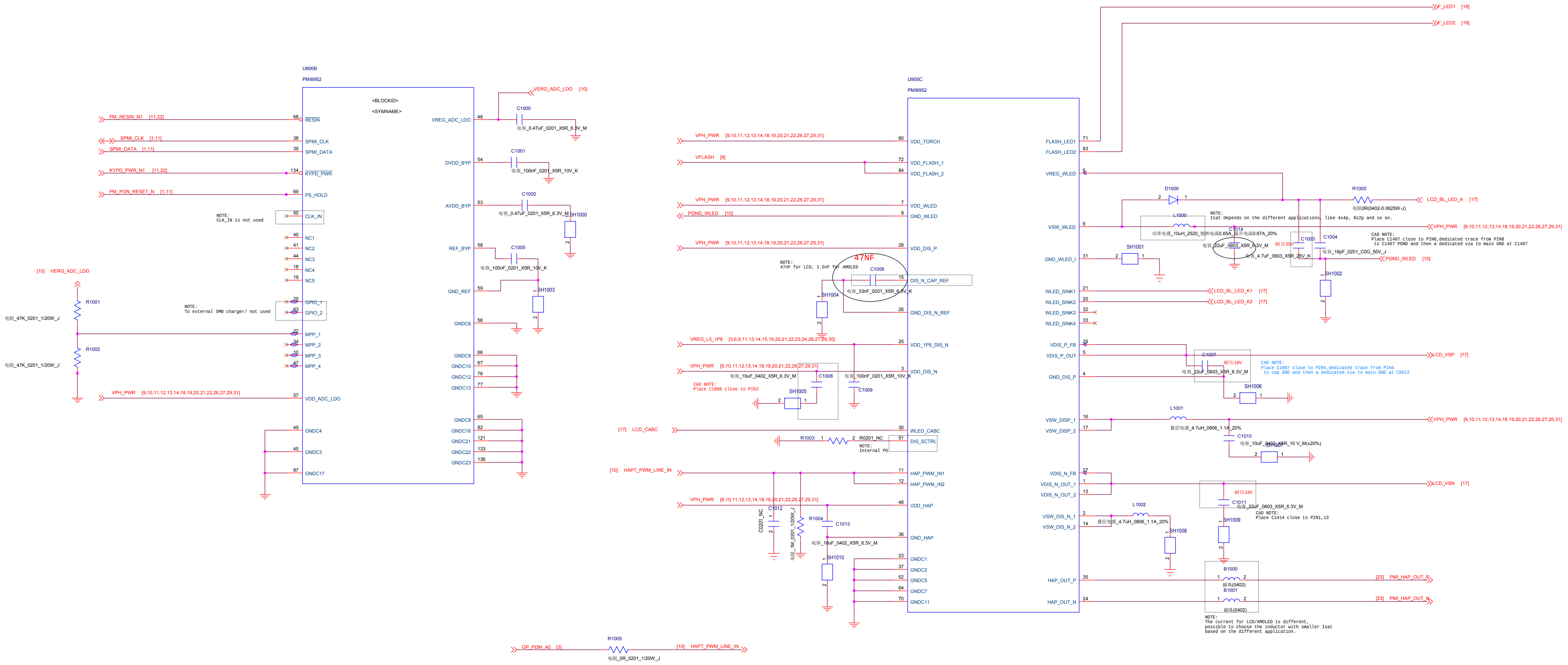


Title	E7_MB_V1		
Size D	Document Number <Doc>	MSM8953 GND	Rev V1.0
Date:	Wednesday, August 09, 2017	Sheet 8 of 34	

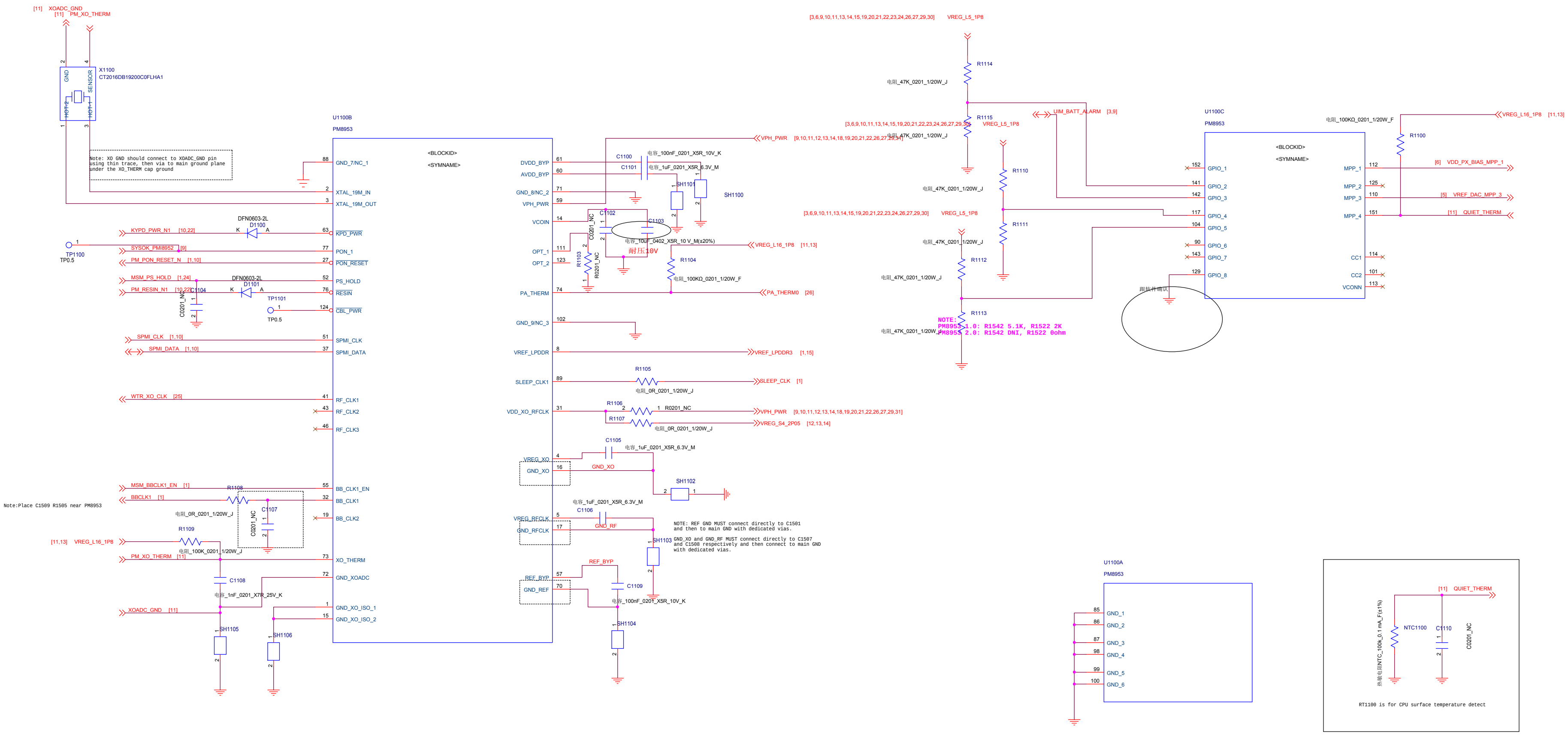
PMI PIN	connection on device without Wipower	connection on device with Wipower
CHG_OK	Pull up to VDD_CAP with 51kohm	Stark PRU CHG_OK
DVI2_EN	Pull down to GND with 0ohm	Stark PRU DIV2_EN
GPIO2	No Connect For parallel charging: Connects to SMB EN pin Do NOT pull up to VDD_CAP	10Kohm pull down to gnd For parallel charging: Connects to SMB EN pin. Do NOT pull up to VDD_CAP



Title	E7_MB_V1.0		
Size D	Document Number <Doc> PMI8952 Charger		Rev V1.0
Date: Wednesday, August 09, 2017	Sheet 9 of 34		



Title	E7_MB_V1.0	
Size D	Document Number <Doc> PMI8952 Control/Interface	Rev V1.0
Date: Wednesday, August 09, 2017	Sheet 10 of 34	



Title: E7_MB_V1

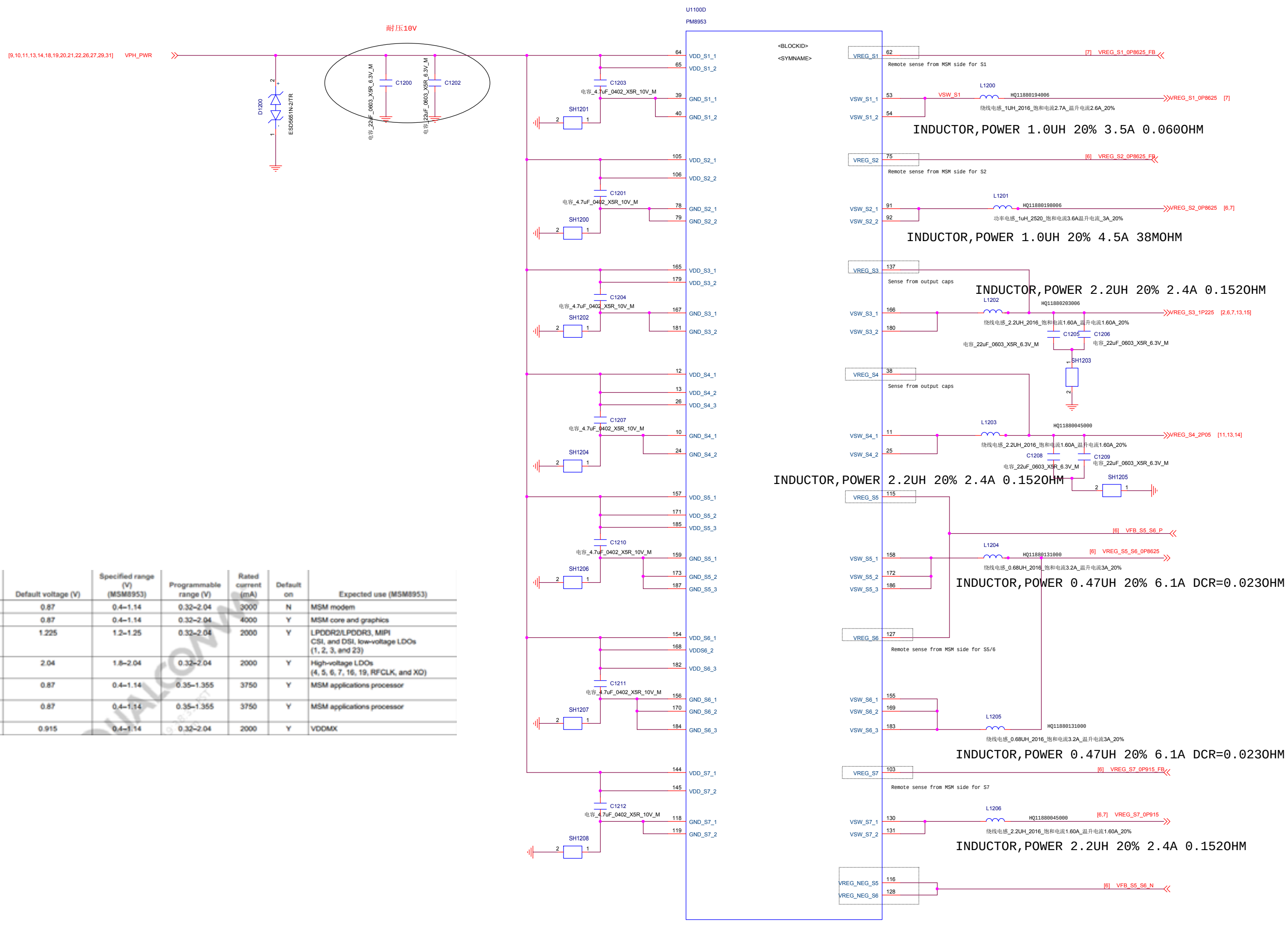
Size: D

Document Number: <Doc> PM8953 Control/Interface

Rev: V1.0

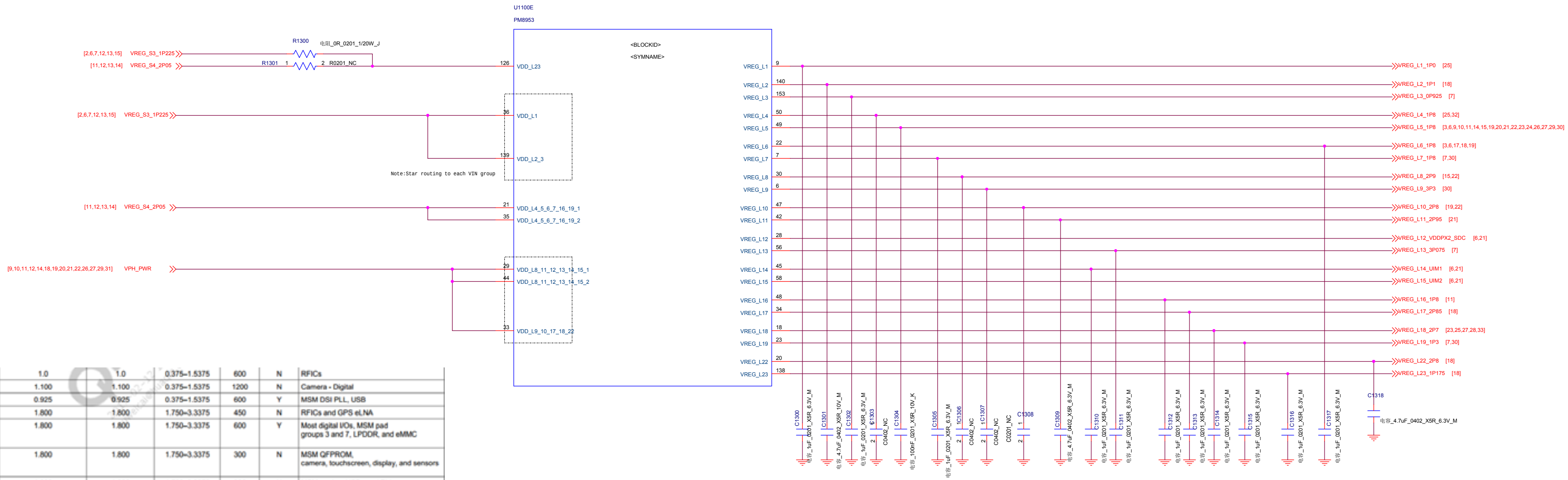
Date: Wednesday, August 09, 2017

Sheet 11 of 34



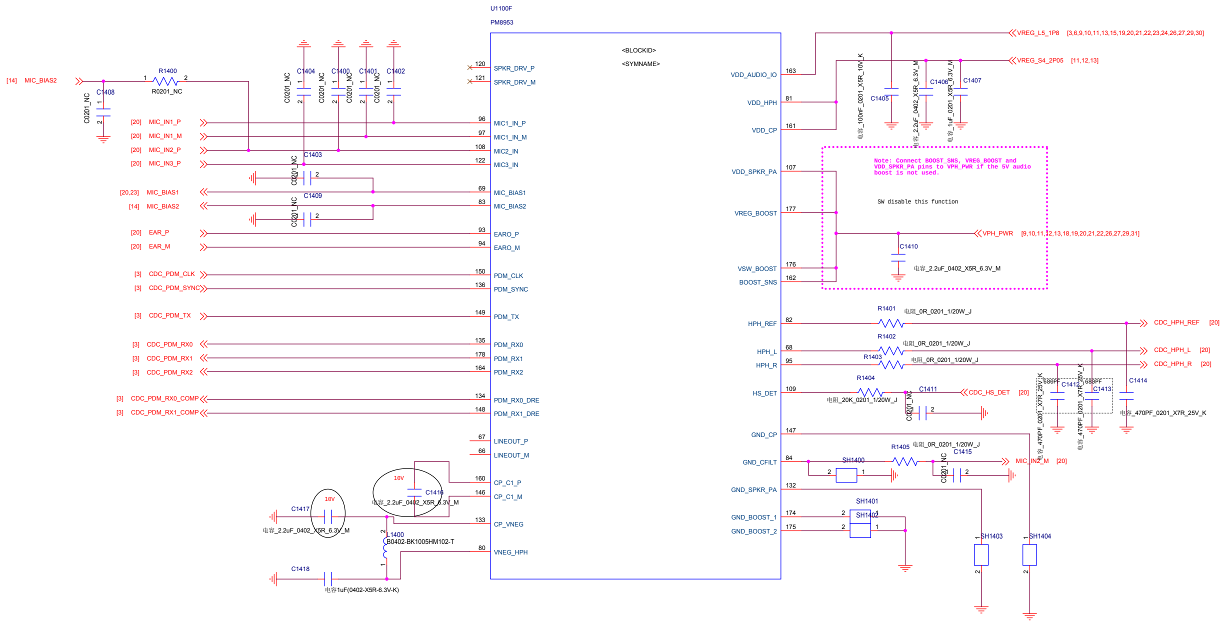
Function	Circuit type	Default voltage (V)	Specified range (V) (MSM8953)	Programmable range (V)	Rated current (mA)	Default on	Expected use (MSM8953)
S1	SMPS	0.87	0.4–1.14	0.32–2.04	3000	N	MSM modem
S2	SMPS	0.87	0.4–1.14	0.32–2.04	4000	Y	MSM core and graphics
S3	SMPS	1.225	1.2–1.25	0.32–2.04	2000	Y	LPDDR2/LPDDR3, MIPI CSI, and DSI, low-voltage LDOs (1, 2, 3, and 23)
S4	SMPS	2.04	1.8–2.04	0.32–2.04	2000	Y	High-voltage LDOs (4, 5, 6, 7, 16, 19, RFCLK, and XO)
S5	SMPS	0.87	0.4–1.14	0.35–1.355	3750	Y	MSM applications processor
S6	SMPS	0.87	0.4–1.14	0.35–1.355	3750	Y	MSM applications processor
S7	SMPS	0.915	0.4–1.14	0.32–2.04	2000	Y	VDDMX

Title: E7_MB_V1		
Size: D	Document Number: <Doc> PM8953 SMPS	Rev: V1.0
Date: Wednesday, August 09, 2017	Sheet 12 of 34	

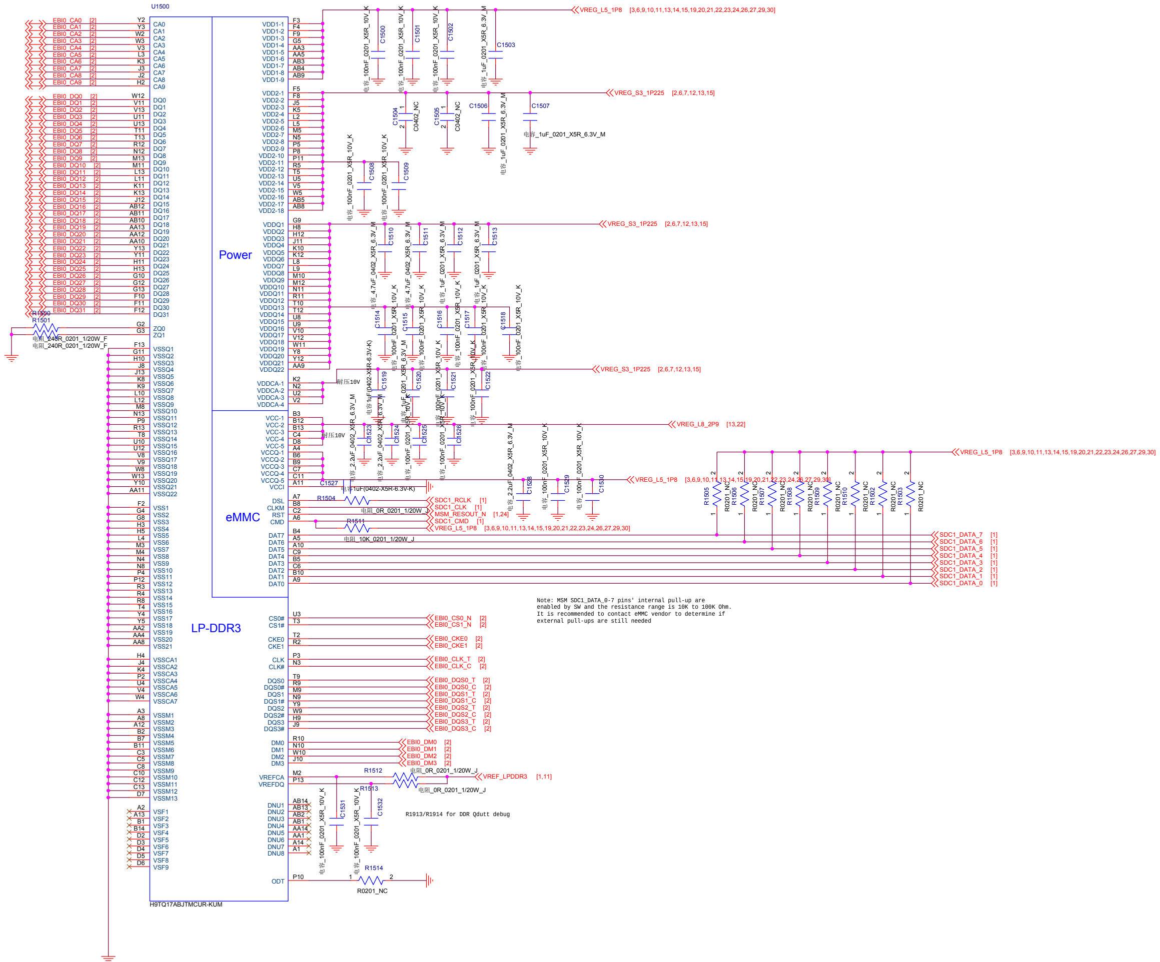


L1	NMOS LDO	1.0	1.0	0.375–1.5375	600	N	RFICs
L2	NMOS LDO	1.100	1.100	0.375–1.5375	1200	N	Camera - Digital
L3	NMOS LDO	0.925	0.925	0.375–1.5375	600	Y	MSM DSI PLL, USB
L4	PMOS LDO	1.800	1.800	1.750–3.3375	450	N	RFICs and GPS eLNA
L5	PMOS LDO	1.800	1.800	1.750–3.3375	600	Y	Most digital I/Os, MSM pad groups 3 and 7, LPDDR, and eMMC
L6	PMOS LDO	1.800	1.800	1.750–3.3375	300	N	MSM QFPROM, camera, touchscreen, display, and sensors
L7	PMOS LDO	1.800	1.800	1.750–3.3375	300	Y	MSM analog, USB, and PLLs, WCN XO, and PM baseband clock driver
L8	PMOS LDO	2.900	2.900	1.750–3.3375	600	Y	eMMC
L9	PMOS LDO	Vout = 3.3 V for VBAT > 3.575 V; Vout = 3 V for VBAT < 3.575 V	3.000–3.300	1.750–3.3375	600	N	WCN

Function	Circuit type	Default V (V)	Specified range (V) (MSM8953)	Programmable range (V)	Rated current (mA)	Default ON	Expected use (MSM8953)
L10	PMOS LDO	3.0	3.0	1.750–3.3375	150	N	Sensors and touchscreen
L11	PMOS LDO	2.950	2.950	1.750–3.3375	800	Y	Micro SD
L12	PMOS LDO	2.950	1.800/2.950	1.750–3.3375	50	Y	MSM pad group 2 and SDC2
L13	PMOS LDO	3.125	3.125	1.750–3.3375	150	Y	MSM USB type C and PMIC and external codec audio
L14	PMOS LDO	1.800	1.800/3	1.750–3.3375	50	N	MSM pad group 5, dual-voltage UIM1, and NFC
L15	PMOS LDO	1.800	1.800/3	1.750–3.3375	50	N	MSM pad group 6 and dual-voltage UIM2
L16	PMOS LDO	1.800	1.800	1.750–3.3375	5	N	PMIC HKADC
L17	PMOS LDO	2.850	2.850	1.750–3.3375	300	N	Camera and display
L18	PMOS LDO	2.700	2.700	1.750–3.3375	150	N	QTI RF front-end
L19	NMOS LDO	1.350	1.350	0.375–1.5375	600	N	MSM analog, WCN, and WGR
L20	Low-noise LDO	1.74	1.74	1.74–3.3375	5	Y	PMIC XO circuits
L21	Low-noise LDO	1.74	1.74	1.74–3.3375	5	N	PMIC RF clock buffers
L22	PMOS LDO	2.800	2.800	1.750–3.3375	150	N	Camera – analog
L23	NMOS LDO	1.15	1.15	0.375–1.5375	600	N	Camera – digital



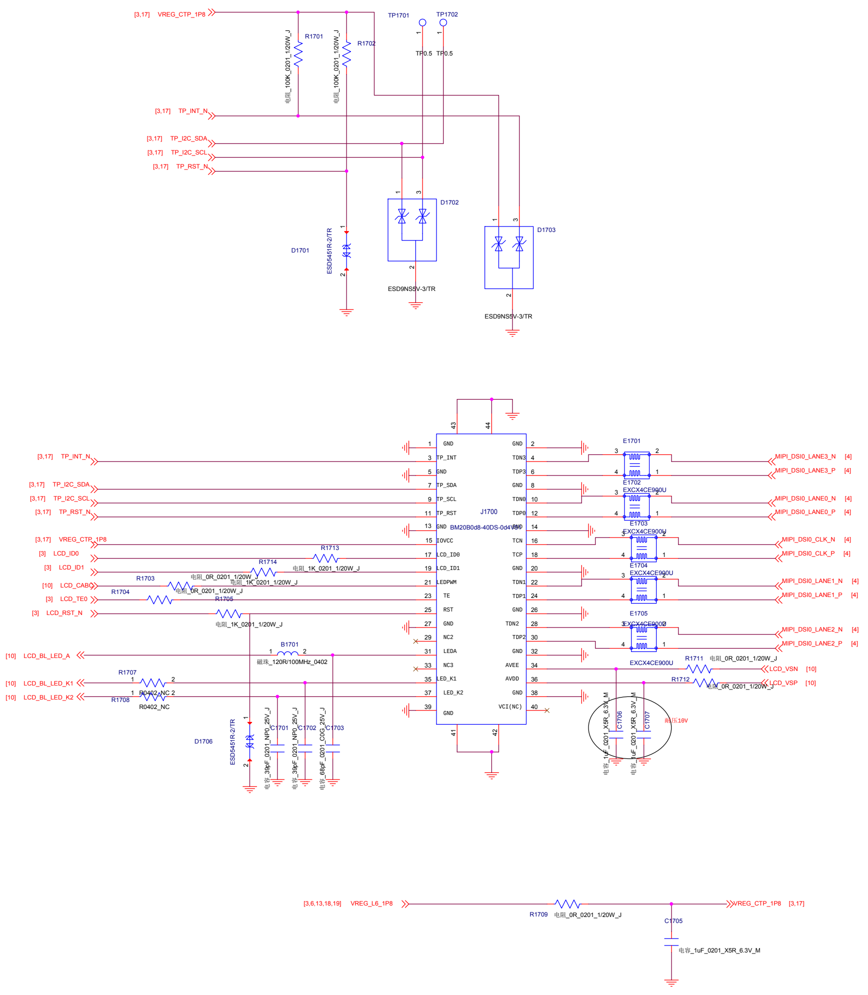
Title	E7_MB_V1	
Size D	Document Number <Doc> PM8953 CODEC/Audio PA	Rev V1.0
Date: Wednesday, August 09, 2017	Sheet 14 of 34	



Title	E7_MB_V1
Size	D
Document Number	<Doc> MEMORY(LPDDR3+EMMC V5.1)
Rev	V1.0
Date:	Wednesday, August 09, 2017
Sheet	15 of 34

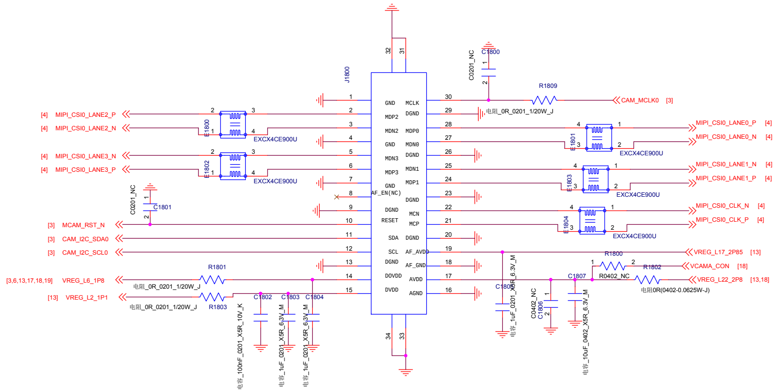
Title: E7_MB_V1		
Size: D	Document Number: <Doc> WCD9326 CODEC	Rev: V1.0
Date: Wednesday, August 09, 2017	Sheet 16 of 34	

Note: If best EMI practices are followed for MIPI CSI/DSI signals, there is no need for common mode choke filters. You may choose to have placeholders for common mode depending upon your design constraints.
Extreme care must be taken that no stubs are created by doing so.

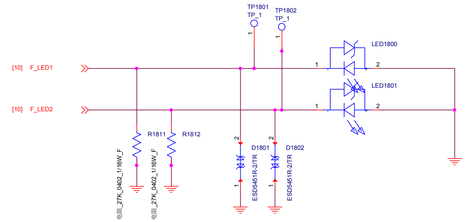


Title	E7_MB_V1		
Size D	Document Number <Doc> LCD/CTP		Rev V1.0
Date: Wednesday, August 09, 2017	Sheet 17 of 34		

Main MONO Camera



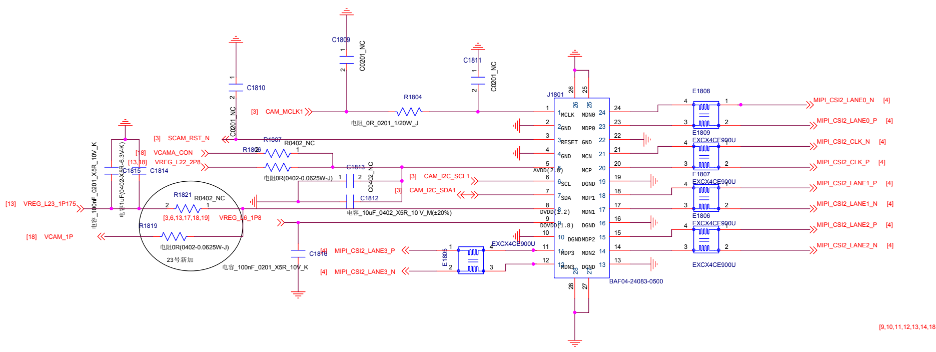
Rear Camera Flash

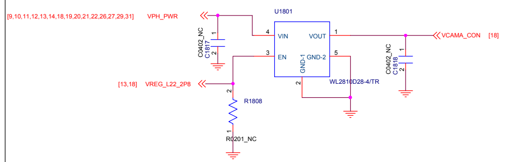


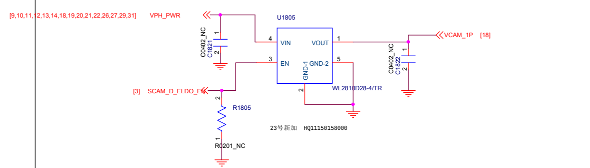
厂家	I2C READ	I2C WRITE	ID
丘钛微	0X 5B	0X 5A	HIGH
舜宇	0X 21	0X 20	LOW

Front Camera

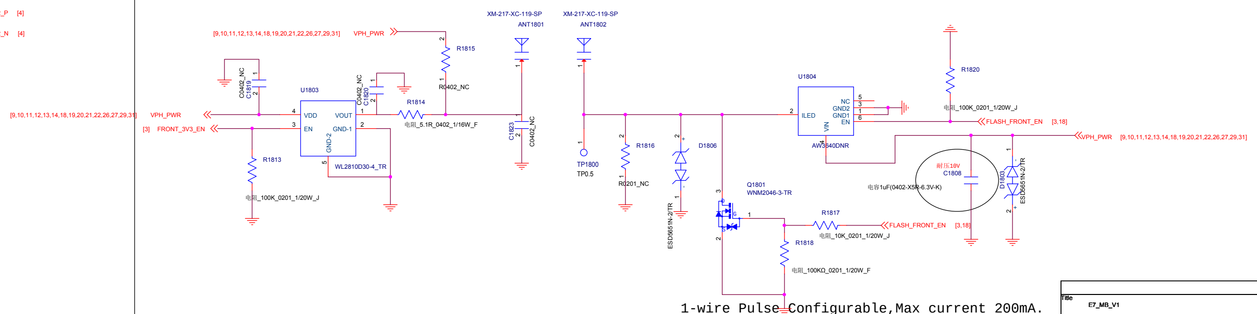
待和结构确认方向







Front Camera Flash

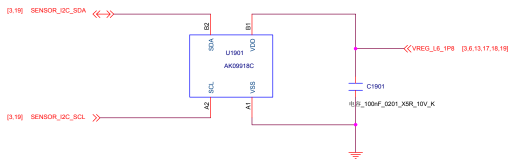


1-wire Pulse Configurable,Max current 200mA.
12.5mA per step .Total 16 STEP.

Title: E7_MB_V1		
Size: D	Document Number: <Doc>	Rev: V1.0
Date: Wednesday, August 09, 2017	Sheet 18 of 34	

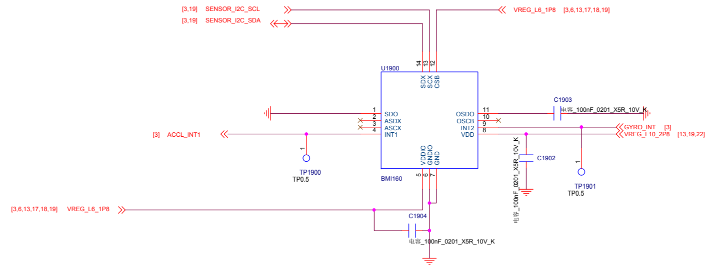
E-compass

厂家	I2C READ	I2C WRITE
BMI120	19	18



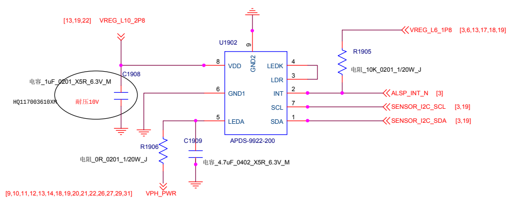
Gyro+A sensor

厂家	I2C READ	I2C WRITE
BMI120	D1	D0

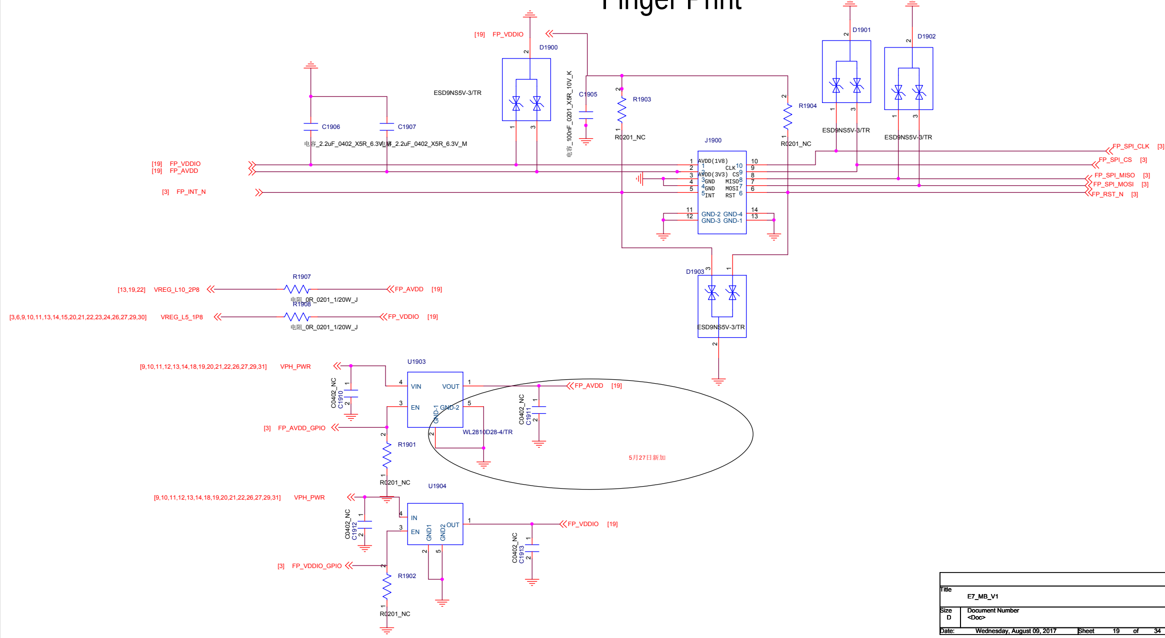


ALS PS

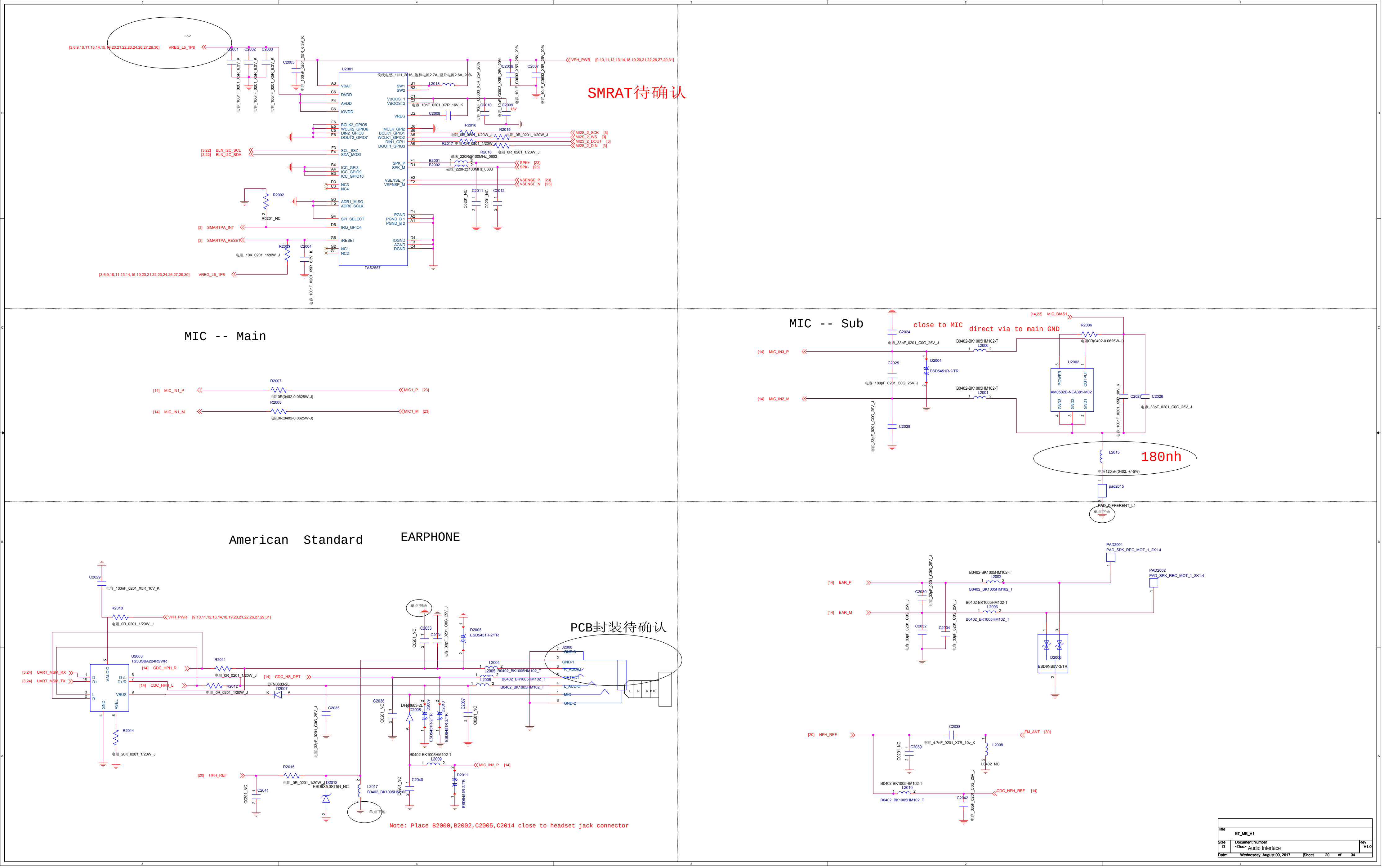
厂家	I2C READ	I2C WRITE
STK3311-35A	91	90

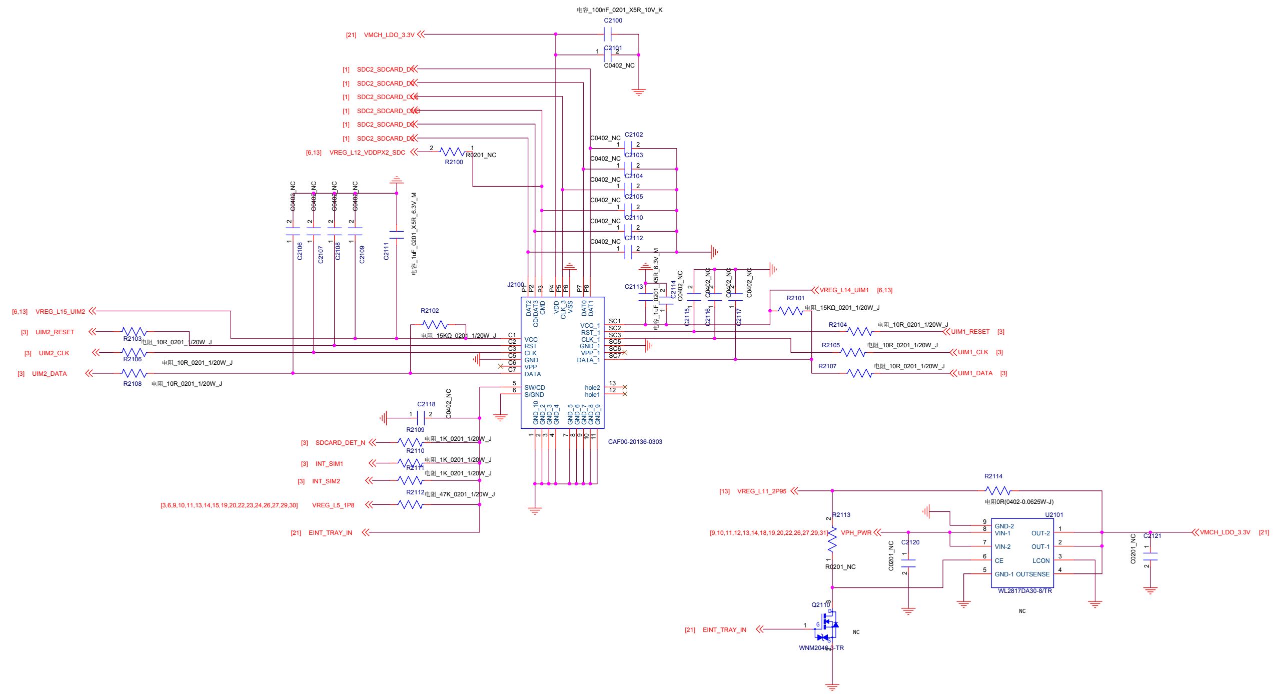


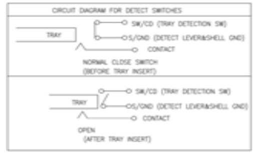
Finger Print



Title	E7_MB_V1
Size D	Document Number <Doc>
Rev	V1.0
Date:	Wednesday, August 09, 2017
Sheet	19 of 34

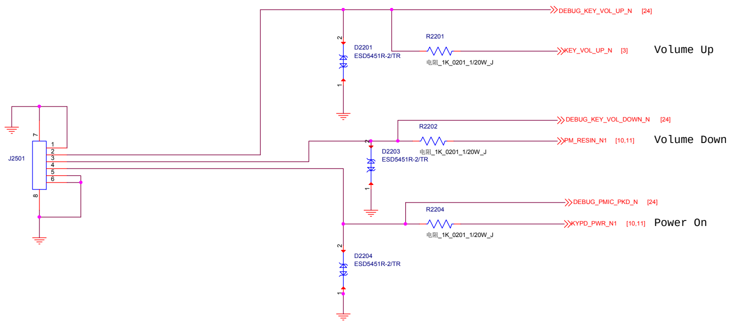

SMRAT待确认
U2001
TAS2557
VREG_L5_1P8
VPH_PWR
MI2S_2_SCK
MI2S_2_WS
MI2S_2_DOUT
MI2S_2_DIN
BLN_I2C_SCL
BLN_I2C_SDA
SPK+
SPK-
VSENSE_P
VSENSE_N
SMARTPA_INT
SMARTPA_RESET
MIC -- Main
MIC_IN1_P
MIC_IN1_M
MIC1_P
MIC1_M
MIC -- Sub
close to MIC
direct via to main GND
MIC_IN3_P
MIC_IN2_M
MIC_BIAS1
U2002
AM0502B-NEA381-M02
180nh
L2015
pad2015
PAD_DIFFERENT_L1
American Standard
EARPHONE
PCB封装待确认
U2003
TS5USBA224RSWR
UART_MSM_RX
UART_MSM_TX
CDC_HPH_R
CDC_HPH_L
CDC_HS_DET
J2000
HPH_REF
MIC_IN2_P
EAR_P
EAR_M
PAD2001
PAD2002
PAD_SPK_REC_MOT_1_2X1.4
D2006
ESD9NS5V-3/TR
FM_ANT
Note: Place B2000,B2002,C2005,C2014 close to headset jack connector
Title
E7_MB_V1
Size
D
Document Number
<Doc> Audio Interface
Rev
V1.0
Date: Wednesday, August 09, 2017
Sheet 20 of 34



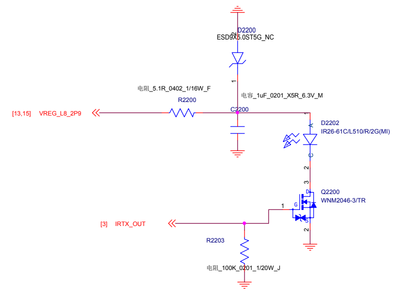


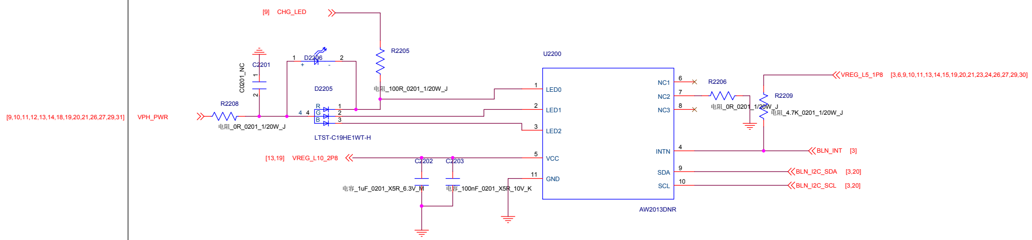
Title	E7_MB_V1	
Size D	Document Number <Doc> SIM/TF card	Rev V1.0
Date: Wednesday, August 09, 2017	Sheet 21 of 34	

Keys

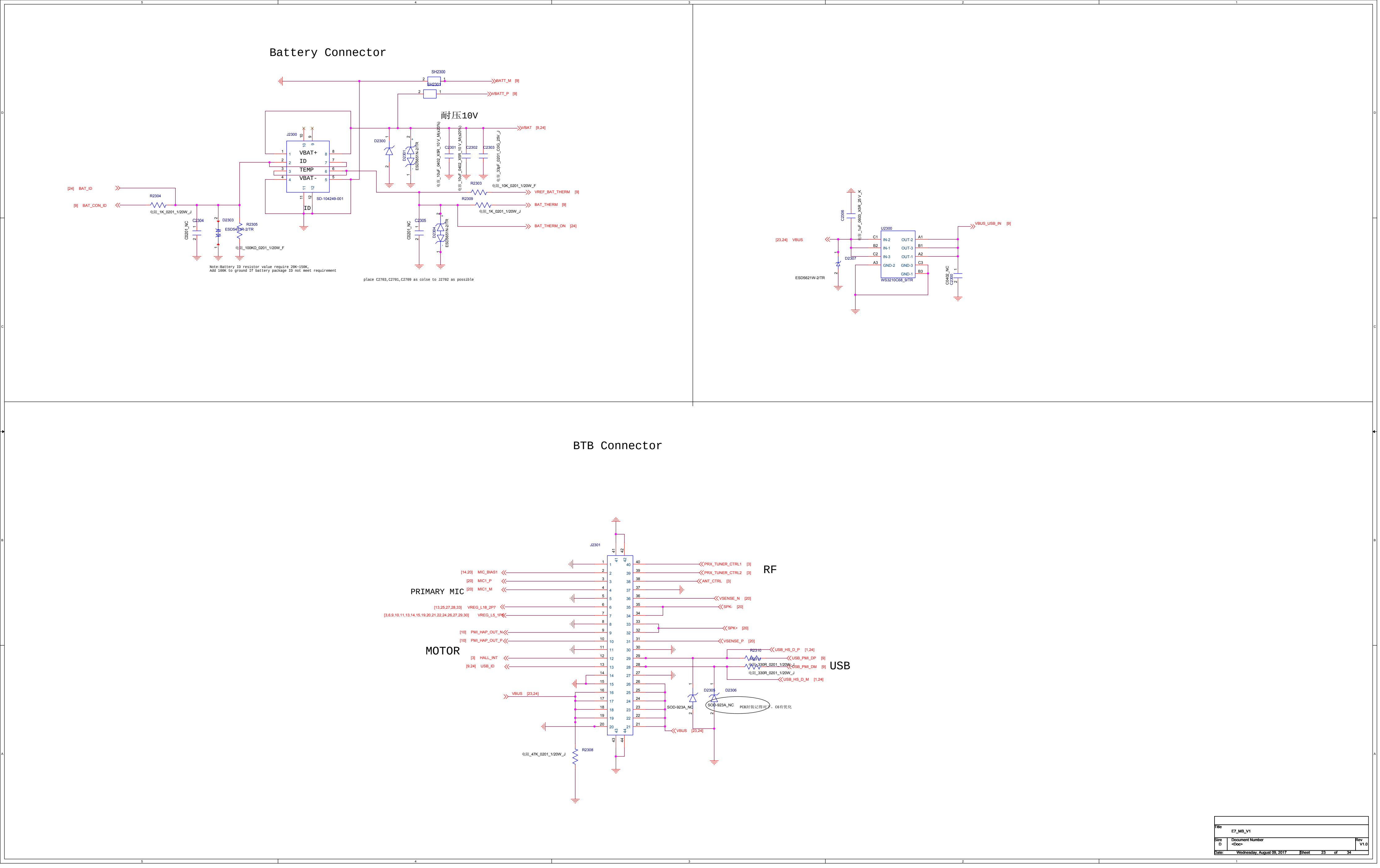


Signal	Description
KPSNS0	Volume Up
PM_RST_N	Volume Down
KYPD_PWR_N	PWER_ON



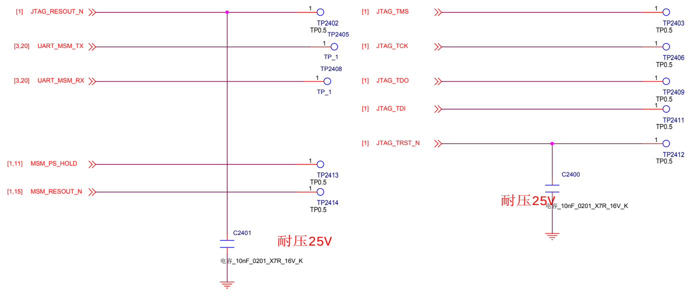


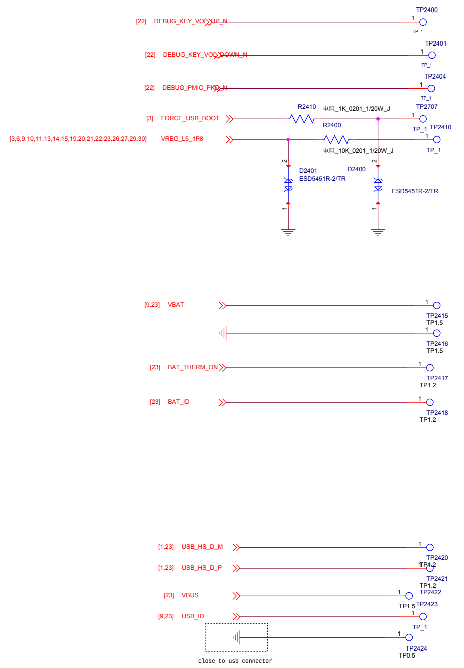
Title	E7_MB_V1		
Size: D	Document Number: <Doc> MISC(Keypad/LED/indicator)		Rev: V1.0
Date: Wednesday, August 09, 2017	Sheet 22 of 34		



Battery Connector
SH2300
BATT_M [9]
SH2301
VBATT_P [9]
耐压10V
VBAT [9,24]
J2300
VBAT+
ID
TEMP
VBAT-
SD-104249-001
D2300
D2301
ESD5651N-2/TR
C2301
C2302
C2303
电容_10uF_0402_X5R_10 V_M(±20%)
电容_33pF_0201_C0G_25V_J
R2303
电阻_10K_0201_1/20W_F
VREF_BAT_THERM [9]
R2309
BAT_THERM [9]
电阻_1K_0201_1/20W_J
BAT_THERM_ON [24]
C2305
C0201_NC
D2304
[24] BAT_ID
[9] BAT_CON_ID
R2304
电阻_1K_0201_1/20W_J
C2304
C0201_NC
D2303
ESD5451R-2/TR
R2305
电阻_100KΩ_0201_1/20W_F
Note:Battery ID resistor value require 20K-150K,
Add 100K to ground If battery package ID not meet requirement
place C2703,C2701,C2709 as colse to J2702 as possible
C2306
电容_1uF_0603_X5R_25 V_K
[23,24] VBUS
D2307
ESD5621W-2/TR
U2300
IN-2
IN-1
IN-3
GND-2
OUT-2
OUT-3
OUT-1
GND-3
GND-1
WS3210C68_8/TR
VBUS_USB_IN [9]
C2300
C0402_NC
BTB Connector
J2301
[14,20] MIC_BIAS1
[20] MIC1_P
[20] MIC1_M
PRIMARY MIC
[13,25,27,28,33] VREG_L18_2P7
[3,6,9,10,11,13,14,15,19,20,21,22,24,26,27,29,30] VREG_L5_1P8
[10] PMI_HAP_OUT_N
[10] PMI_HAP_OUT_P
MOTOR
[3] HALL_INT
[9,24] USB_ID
VBUS [23,24]
PRX_TUNER_CTRL1 [3]
PRX_TUNER_CTRL2 [3]
ANT_CTRL [3]
RF
VSENSE_N [20]
SPK- [20]
SPK+ [20]
VSENSE_P [20]
USB_HS_D_P [1,24]
R2310
USB_PMI_DP [9]
USB_PMI_DM [9]
电阻_330R_0201_1/20W_J
USB_HS_D_M [1,24]
USB
D2305
D2306
SOD-923A_NC
PCB封装记得对上，C6有优化
VBUS [23,24]
R2308
电阻_47K_0201_1/20W_J
Title
E7_MB_V1
Size D
Document Number <Doc>
Rev V1.0
Date: Wednesday, August 09, 2017
Sheet 23 of 34

JTAG CONNECTOR





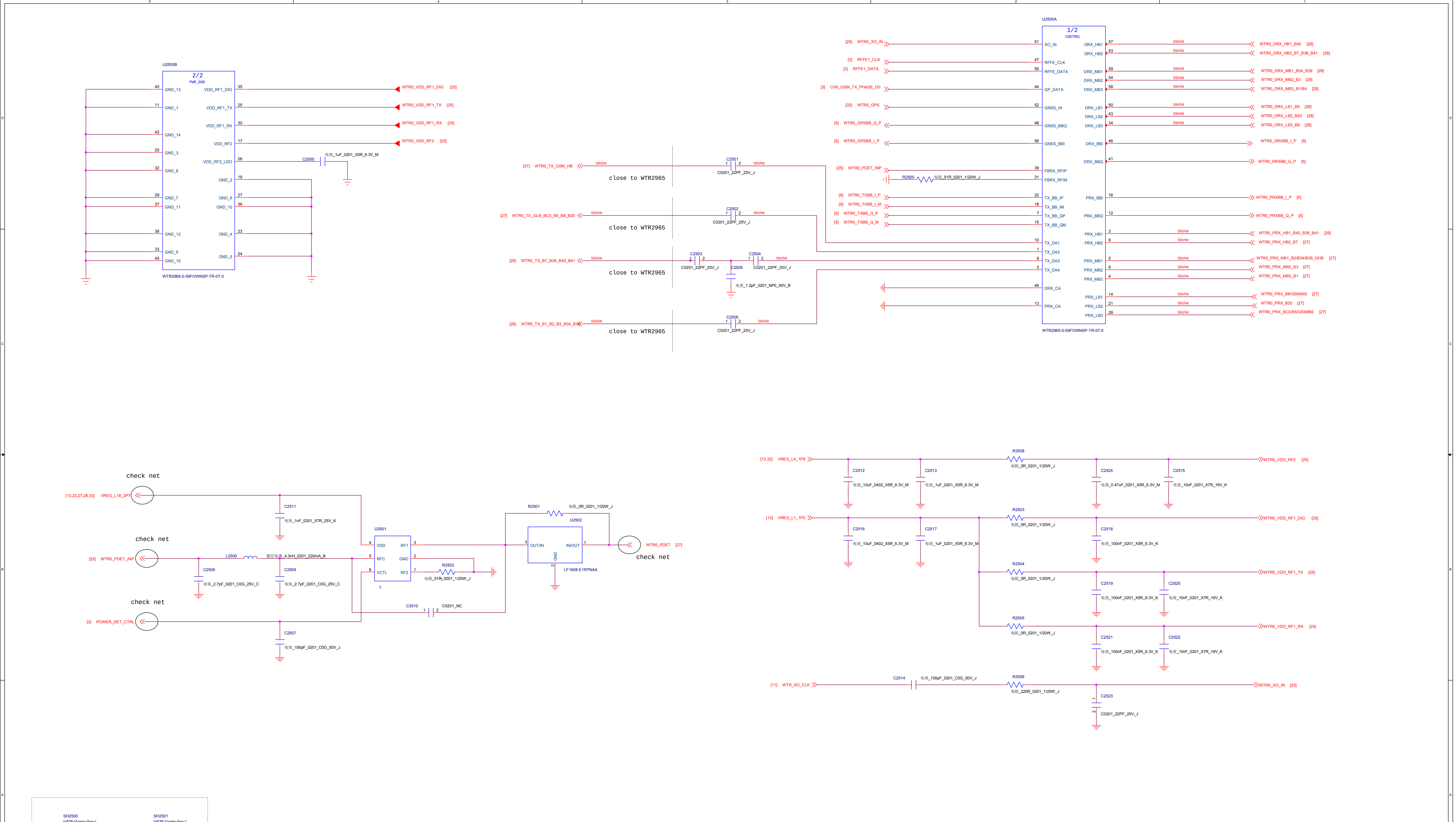
MARK POINT

PCBA AUTOMATIC TEST point

Grounding

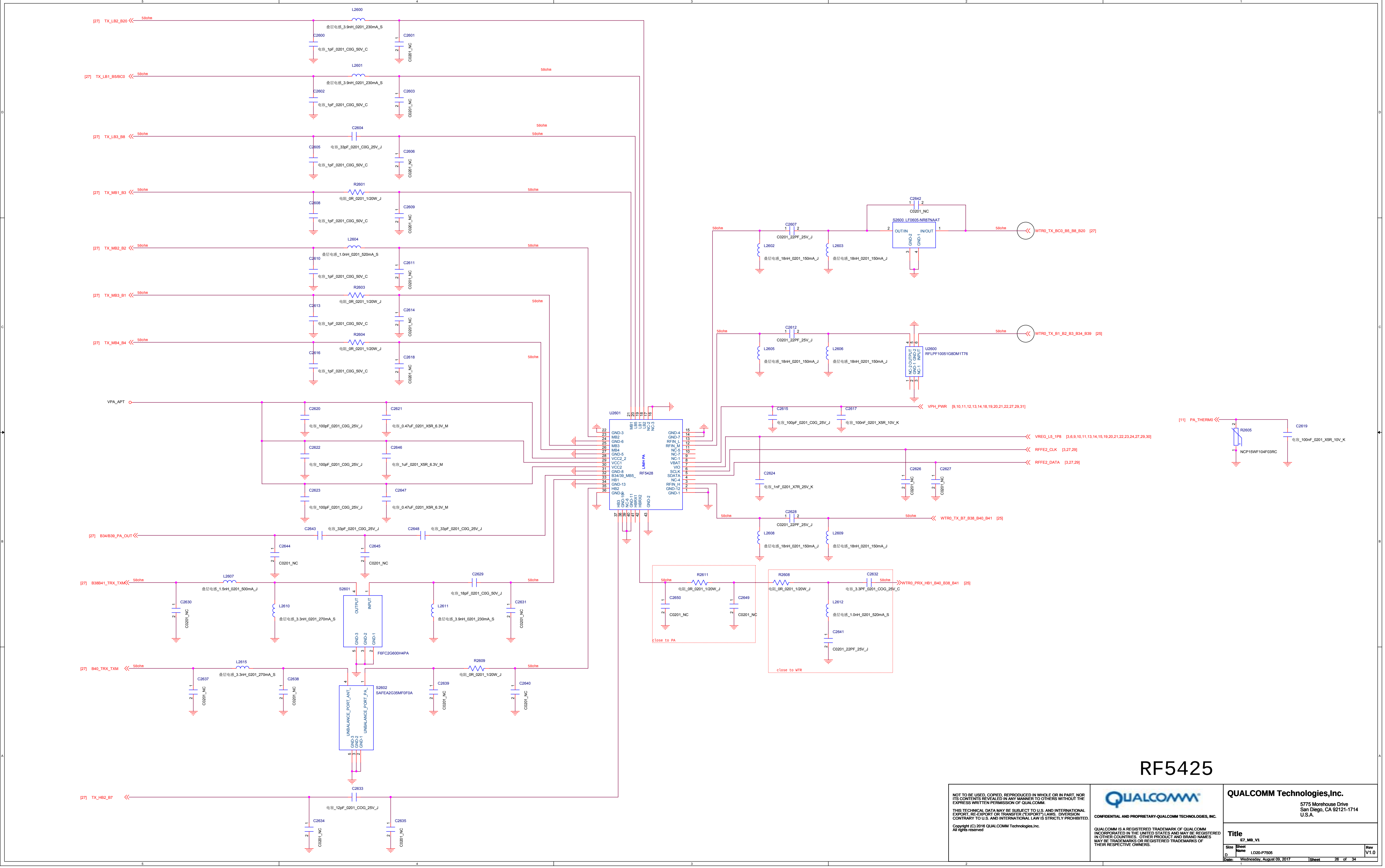
Shielding

Title	E7_MB_V1	
Size D	Document Number <Doc> TP/GND/Shielding	Rev V1.0
Date: Wednesday, August 09, 2017	Sheet 24 of 34	



QUALCOMM Technologies,Inc.
5775 Morehouse Drive
San Diego, CA 92121-1714
U.S.A.

Title	E7_MB_V1		
Size D	Sheet Name LD20-P7505		Rev V1.0
Date: Wednesday, August 09, 2017		Sheet 25 of 34	



RF5425

NOT TO BE USED, COPIED, REPRODUCED IN WHOLE OR IN PART, NOR ITS CONTENTS REVEALED IN ANY MANNER TO OTHERS WITHOUT THE EXPRESS WRITTEN PERMISSION OF QUALCOMM.

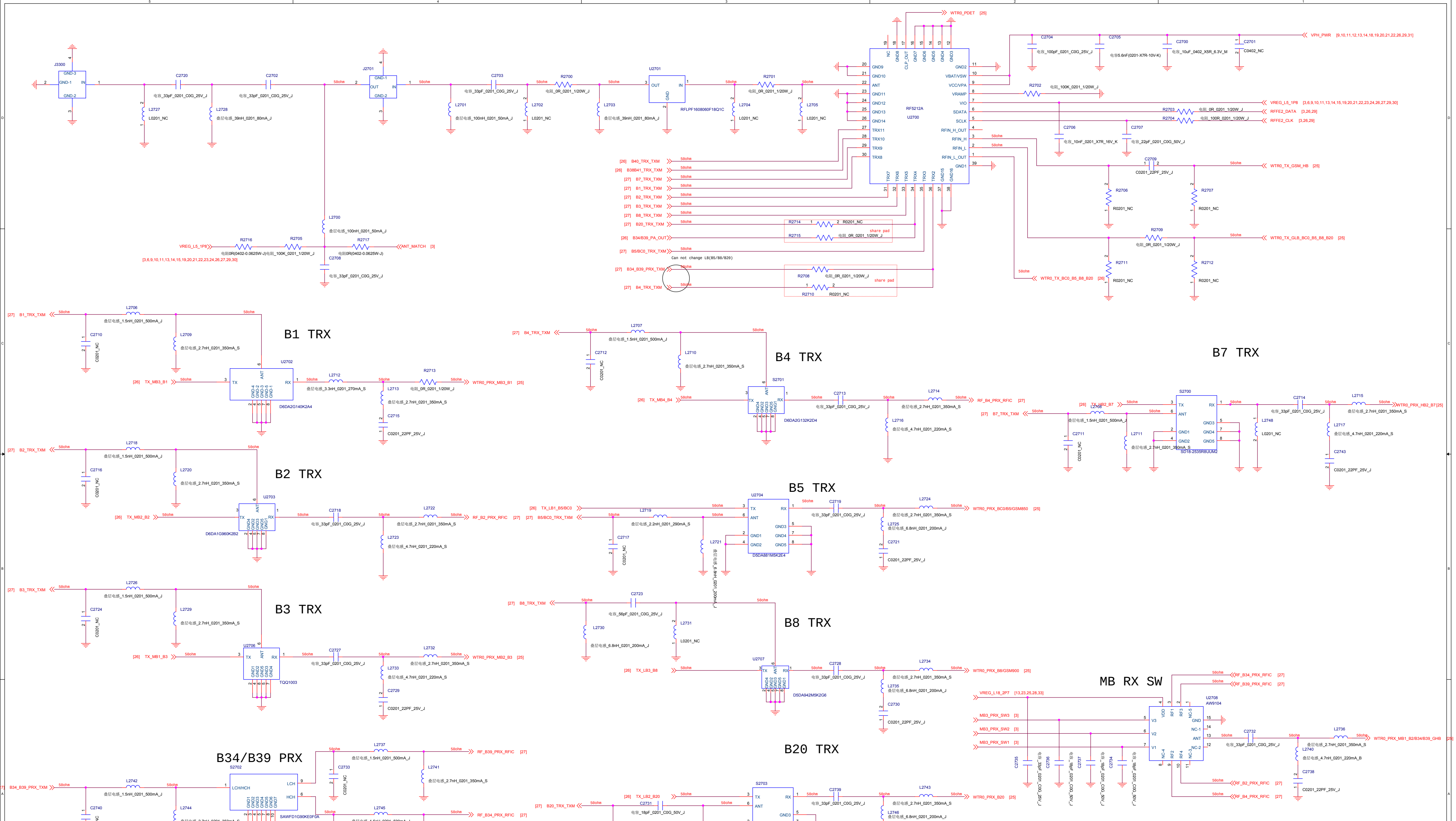
THIS TECHNICAL DATA MAY BE SUBJECT TO U.S. AND INTERNATIONAL EXPORT, RE-EXPORT OR TRANSFER ("EXPORT") LAWS. DIVERSION CONTRARY TO U.S. AND INTERNATIONAL LAW IS STRICTLY PROHIBITED.

CONFIDENTIAL AND PROPRIETARY-QUALCOMM TECHNOLOGIES, INC.

QUALCOMM IS A REGISTERED TRADEMARK OF QUALCOMM INCORPORATED IN THE UNITED STATES AND MAY BE REGISTERED IN OTHER COUNTRIES. OTHER PRODUCT AND BRAND NAMES MAY BE TRADEMARKS OR REGISTERED TRADEMARKS OF THEIR RESPECTIVE OWNERS.

QUALCOMM Technologies,Inc.
5775 Morehouse Drive
San Diego, CA 92121-1714
U.S.A.

Title	E7_MB_V1	
Size D	Sheet Name LD20-P7505	Rev V1.0
Date: Wednesday, August 09, 2017	Sheet 26 of 34	



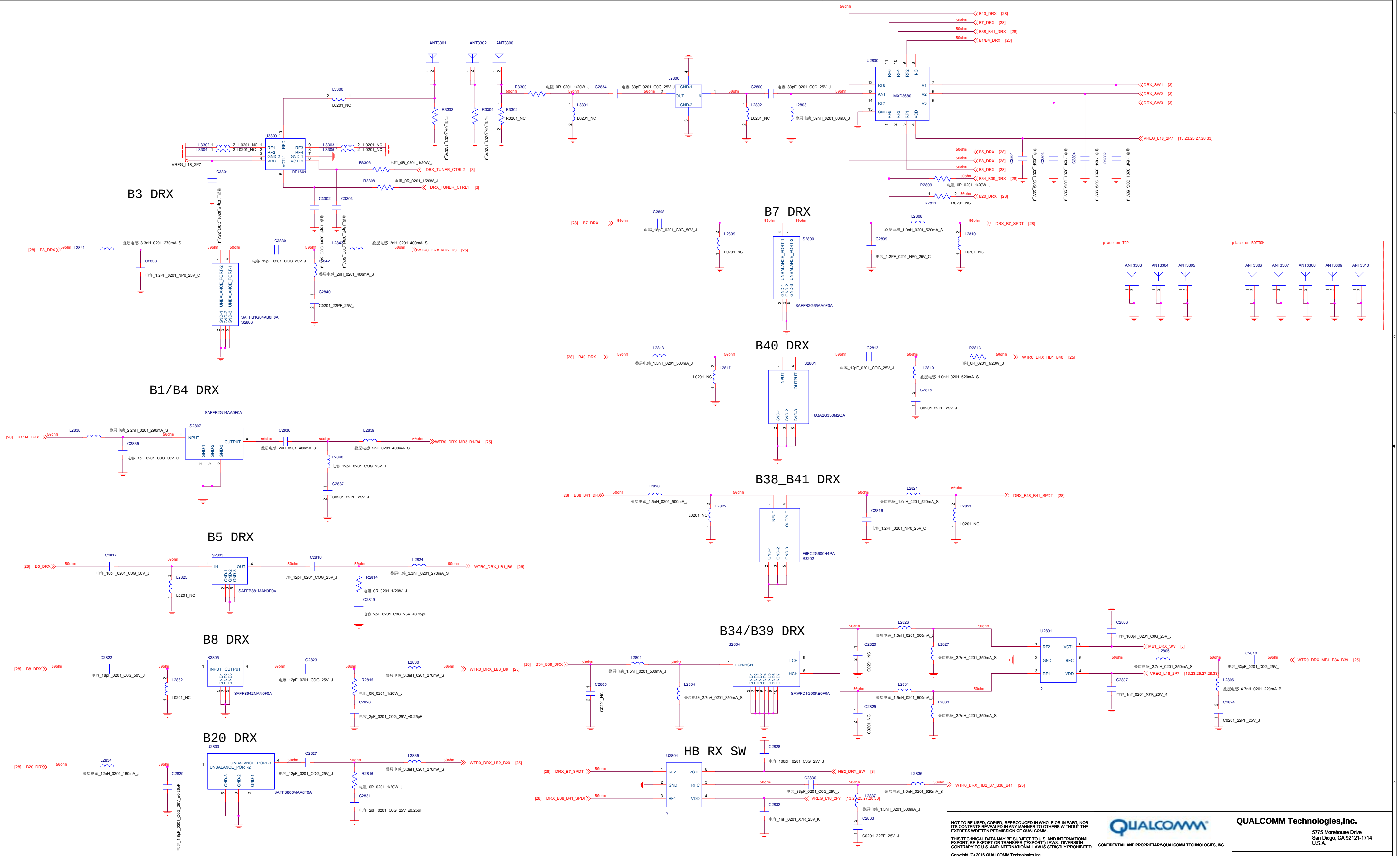
QUALCOMM Technologies,Inc.

5775 Morehouse Drive
San Diego, CA 92121-1714
U.S.A.

Title
E7_MB_V1

Size	Sheet Name	Rev
D	LD20-P7505	V1.0

Date: Wednesday, August 09, 2017 Sheet 27 of 34

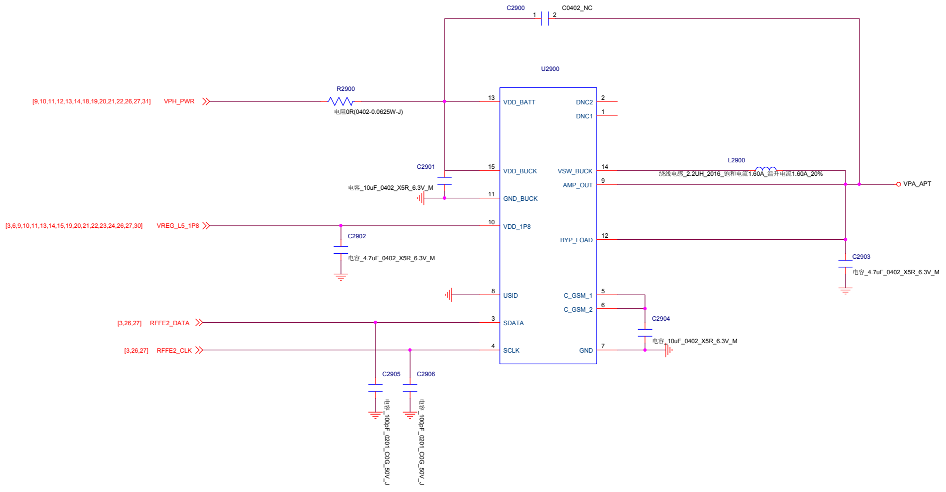


QUALCOMM Technologies,Inc.
5775 Morehouse Drive
San Diego, CA 92121-1714
U.S.A.

Title
E7_MB_V1

Size	Sheet Name	Rev
D	LD20-P7505	V1.0

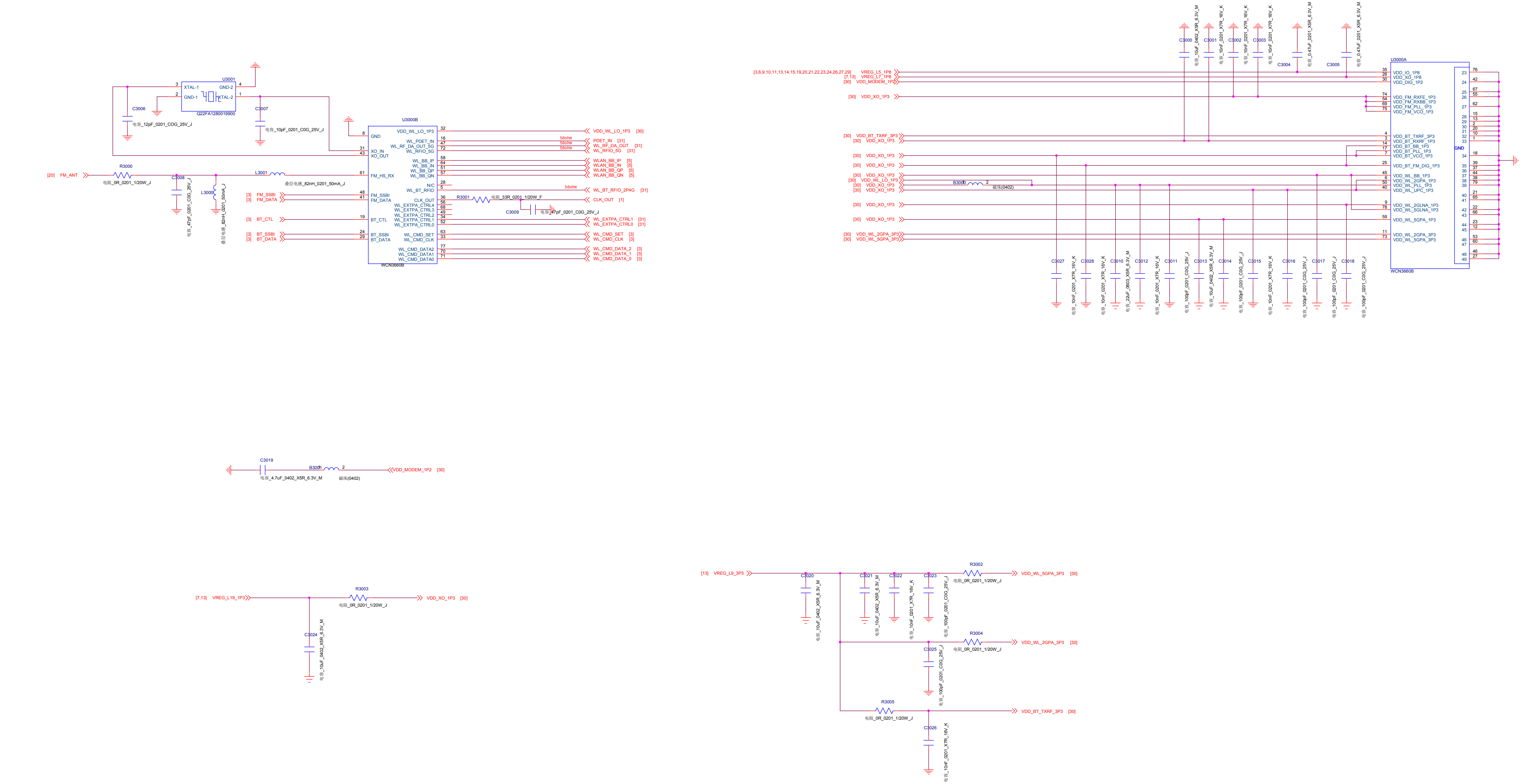
Date: Wednesday, August 09, 2017 Sheet 28 of 34



QFE2101

QUALCOMM Technologies,Inc.
5775 Morehouse Drive
San Diego, CA 92121-1714
U.S.A.

Title E7_MB_V1		
Size D	Sheet Name LD20-P7505	Rev V1.0
Date: Wednesday, August 09, 2017	Sheet 29 of 34	



WCN3680B

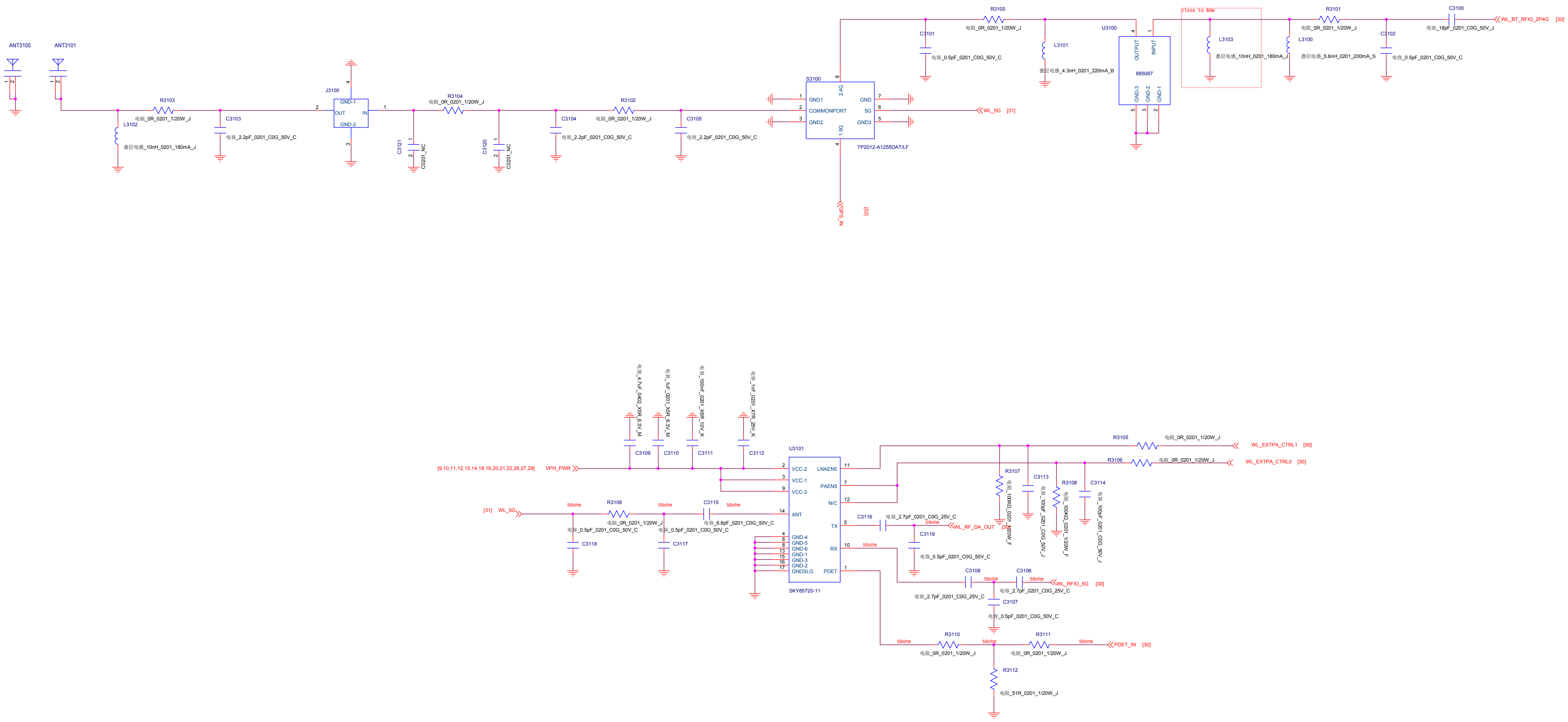
QUALCOMM Technologies,Inc.

5775 Morehouse Drive
San Diego, CA 92121-1714
U.S.A.

Title E7_MB_V1

Size	Sheet Name	Rev
D	LD20-P7505	V1.0

Date: Wednesday, August 09, 2017 — Sheet 30 of 34

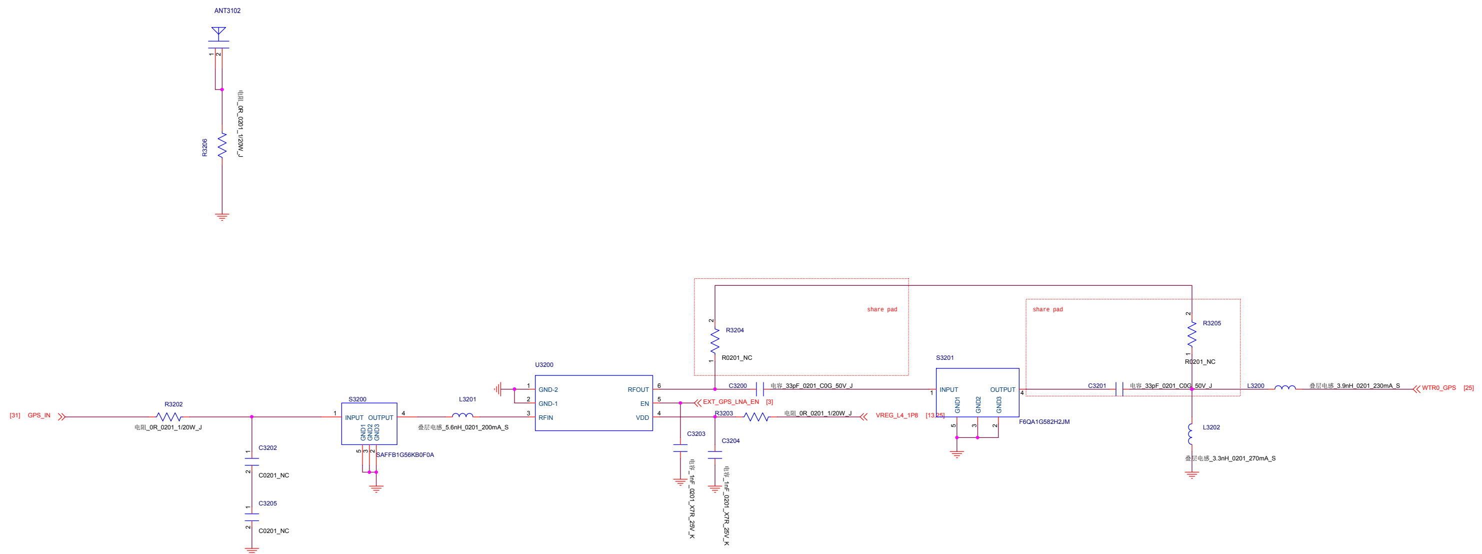


WLAN_FEM

QUALCOMM Technologies,Inc.

5775 Morehouse Drive
San Diego, CA 92121-1714
U.S.A.

Title E7_MB_V1		
Size D	Sheet Name LD20-P7505	Rev V1.0
Date: Wednesday, August 09, 2017	Sheet 31 of 34	



	extra S3201 option	Non-extra S3201 option
R3200	DNI	0ohm
R3201	DNI	0ohm
C3200	33pF	DNI
C3201	33pF	DNI
S3201	SAFFB1G56KB0F0A	DNI

GPS

QUALCOMM Technologies,Inc.
5775 Morehouse Drive
San Diego, CA 92121-1714
U.S.A.

Title
E7_MB_V1

Size D | Sheet Name LD20-P7505 | Rev V1.0

Date: Wednesday, August 09, 2017 | Sheet 32 of 34

Change list

QUALCOMM Technologies,Inc.
5775 Morehouse Drive
San Diego, CA 92121-1714
U.S.A.

Title: E7_MB_V1		
Size: D	Sheet Name: LD20-P7505	Rev: V1.0
Date: Wednesday, August 09, 2017	Sheet 34 of 34	